Introduction to Biomaterials

Lecture [7]

Natural biomaterials

Goals

TO:

- Define natural biomaterials.
- Explain why natural materials are useful in biomedical applications.
- Identify the main requirements of natural biomaterials.
- Classify common natural biomaterials into major groups.
- Give examples of natural biomaterials used in medical devices and tissue engineering.

What Are Natural Biomaterials?

- **Natural biomaterials** :are materials obtained from biological or natural sources and used in biomedical applications.
- They are often interesting because biological materials already perform useful functions inside living systems.
- Natural materials are not random. Their structure and composition are usually related to their biological function.
- **Examples of biological functions:**
  - Cell attachment
  - Cell communication
  - Storage of molecules
  - Mechanical support
  - Tissue repair

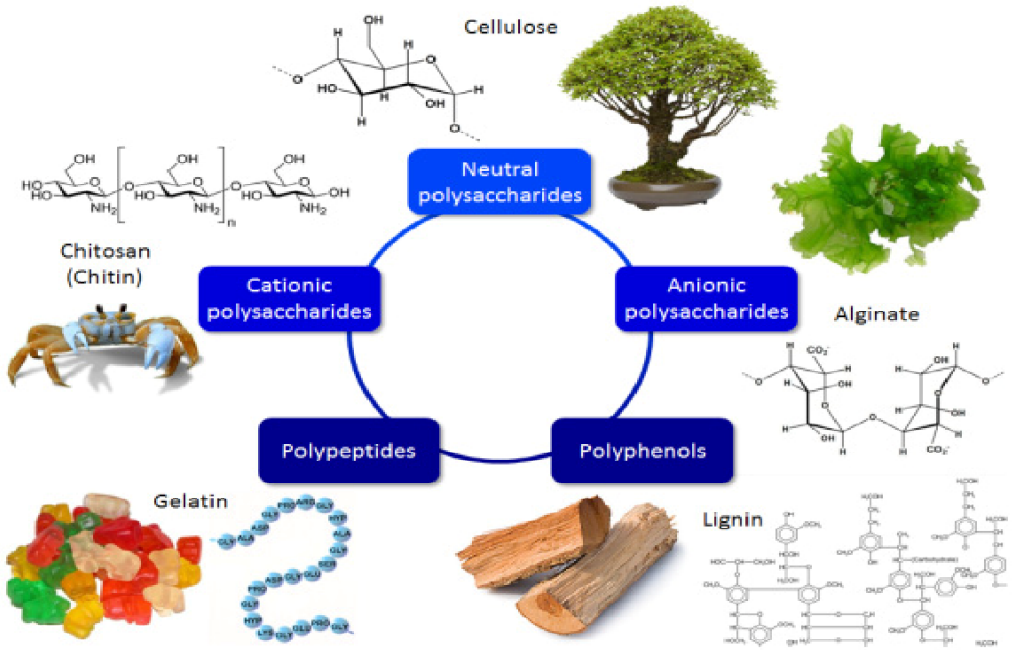
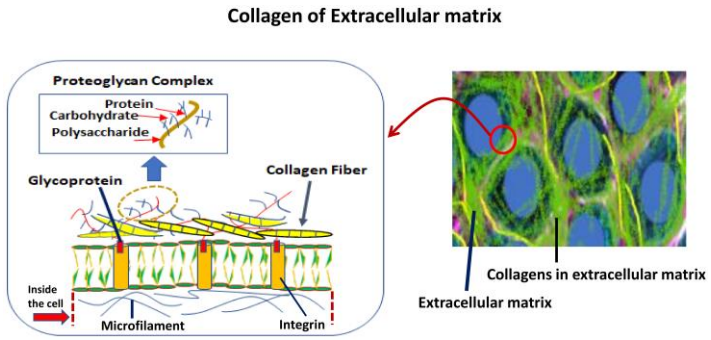


Fig. 1. Natural renewable biomaterials.

Why Are Natural Biomaterials Important?

- Natural biomaterials **are widely studied** because they can **interact well with cells** and **tissues**.
- **Advantages**
  - Support cell attachment and growth
  - May help cell differentiation
  - Can improve the mechanical behavior of synthetic biomaterials
  - Can bind and deliver biological molecules
  - Useful in tissue engineering and regenerative medicine
- **Simple example**
  - A natural polymer such as collagen can provide a familiar environment for cells because collagen is already present in the extracellular matrix.



Requirements for Natural Biomaterials

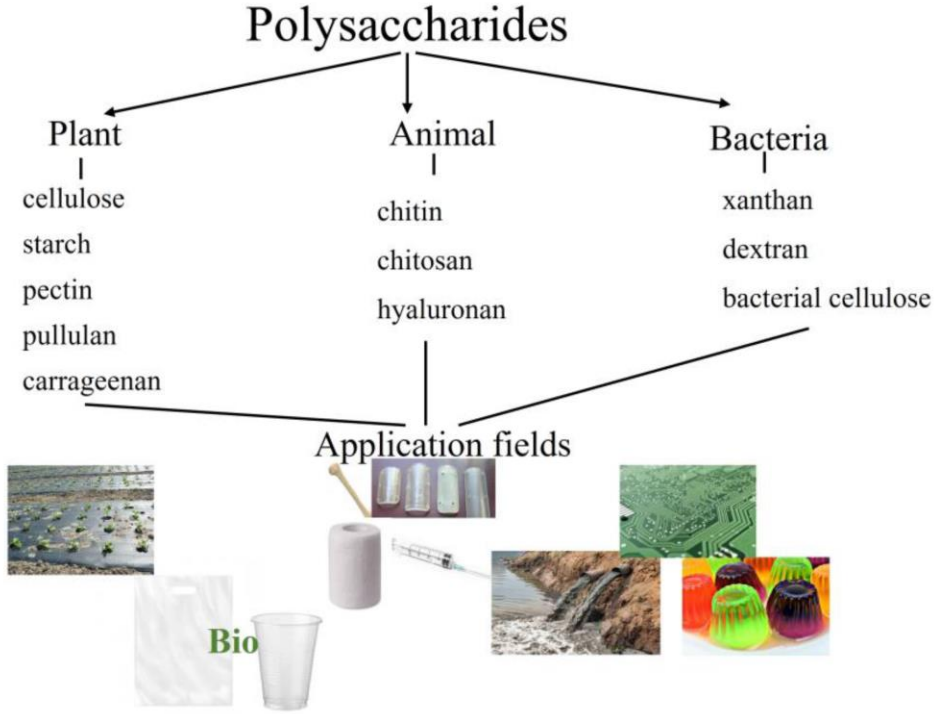
- Like **synthetic** biomaterials, **natural** biomaterials **should** meet important biomedical requirements.
- **A good natural biomaterial should be:**
  - Non-toxic
  - Non-inflammatory
  - Non-allergenic
  - Biocompatible
  - Mechanically suitable for its application
  - Able to support cell attachment when needed
  - Able to support cell differentiation when needed
  - Low cost and available when possible
- Being “natural” does not automatically mean the material is safe. It still needs testing before medical use!!

Main Classes of Natural Biomaterials

- Most common **natural biomaterials** are **polymeric**.
- They are mainly classified into:
- **1. Protein-based natural polymers**
  - *Examples:*
    - Collagen
    - Gelatin
    - Silk fibroin
    - Fibrin
    - Elastin

Main Classes of Natural Biomaterials

- Most common natural biomaterials are polymeric.
- They are mainly classified into:
- 2. Polysaccharide-based natural polymers
  - *Examples:*
    - Chitosan
    - Starch
    - Alginate
    - Hyaluronan
    - Chondroitin sulfate
    - Dextran



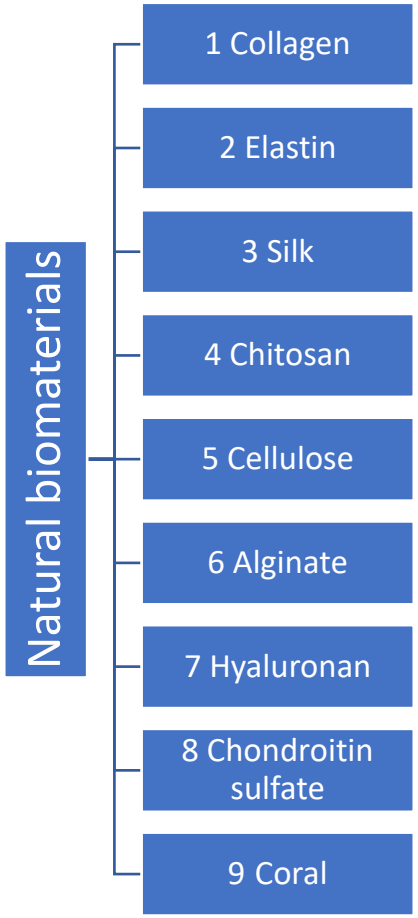
Main Classes of Natural Biomaterials

- They are mainly classified into:
- **3. Mineral-based natural materials**
- Example:
  - Coral-derived calcium carbonate





Natural biomaterials

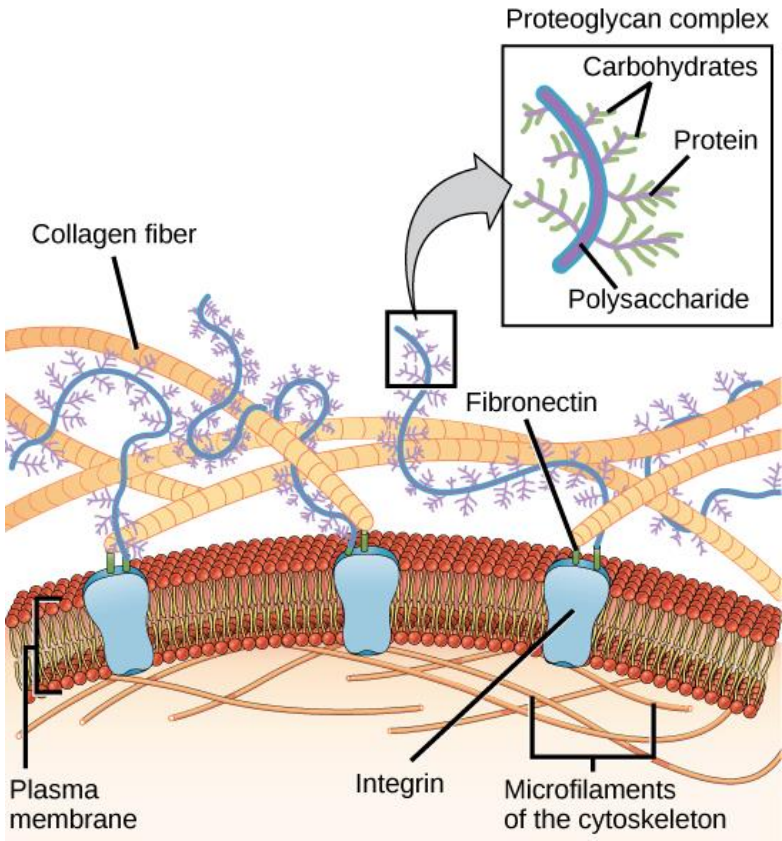


1.Collagen

Collagen



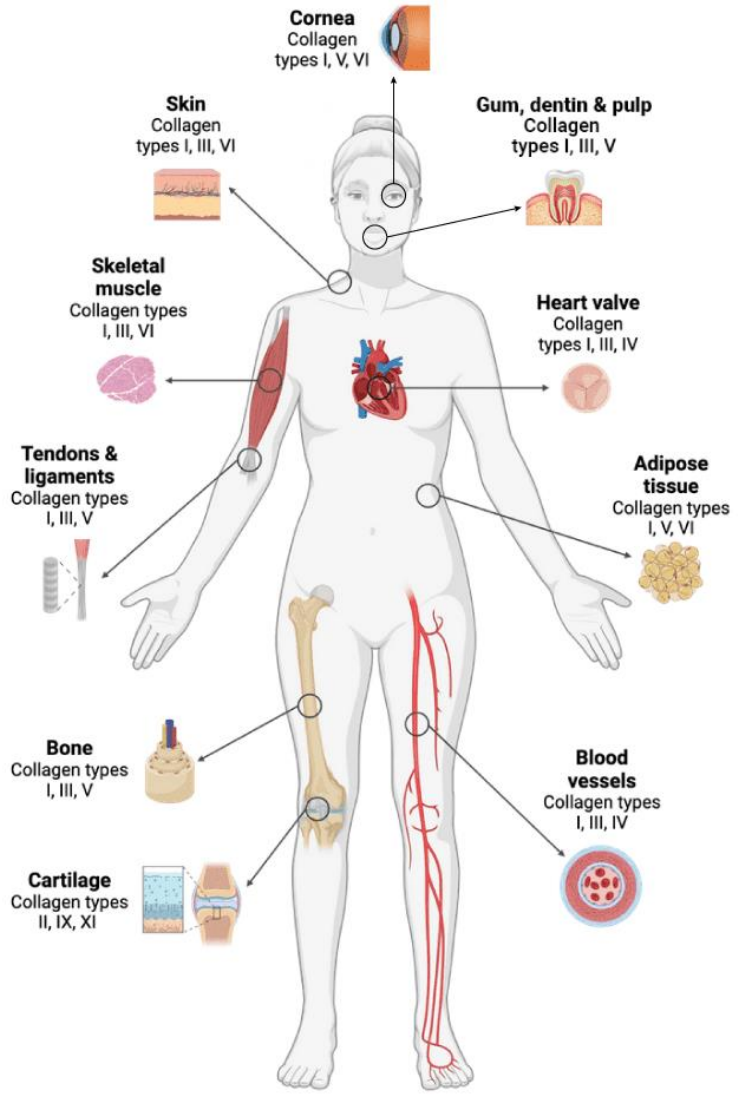
- Collagen is a natural **protein-based biomaterial** found only in animals.
- It is one of the most important structural proteins in the body and is a major component of the **extracellular matrix (ECM)**.
- **Why collagen is important**
  - Provides structural support to tissues
  - Helps maintain tissue strength and integrity
  - Supports cell attachment, migration, and organization
  - Plays an important role in wound healing and tissue regeneration



Collagen in the Human Body

- Collagen is the **most abundant protein in the human body**.
- It makes up about **30% of total body proteins**.
- **Main locations**
  - Skin
  - Bone
  - Tendons
  - Ligaments
  - Blood vessels
  - Cartilage
  - Cornea
  - Connective tissues

The strength and structure of many tissues depend strongly on collagen.

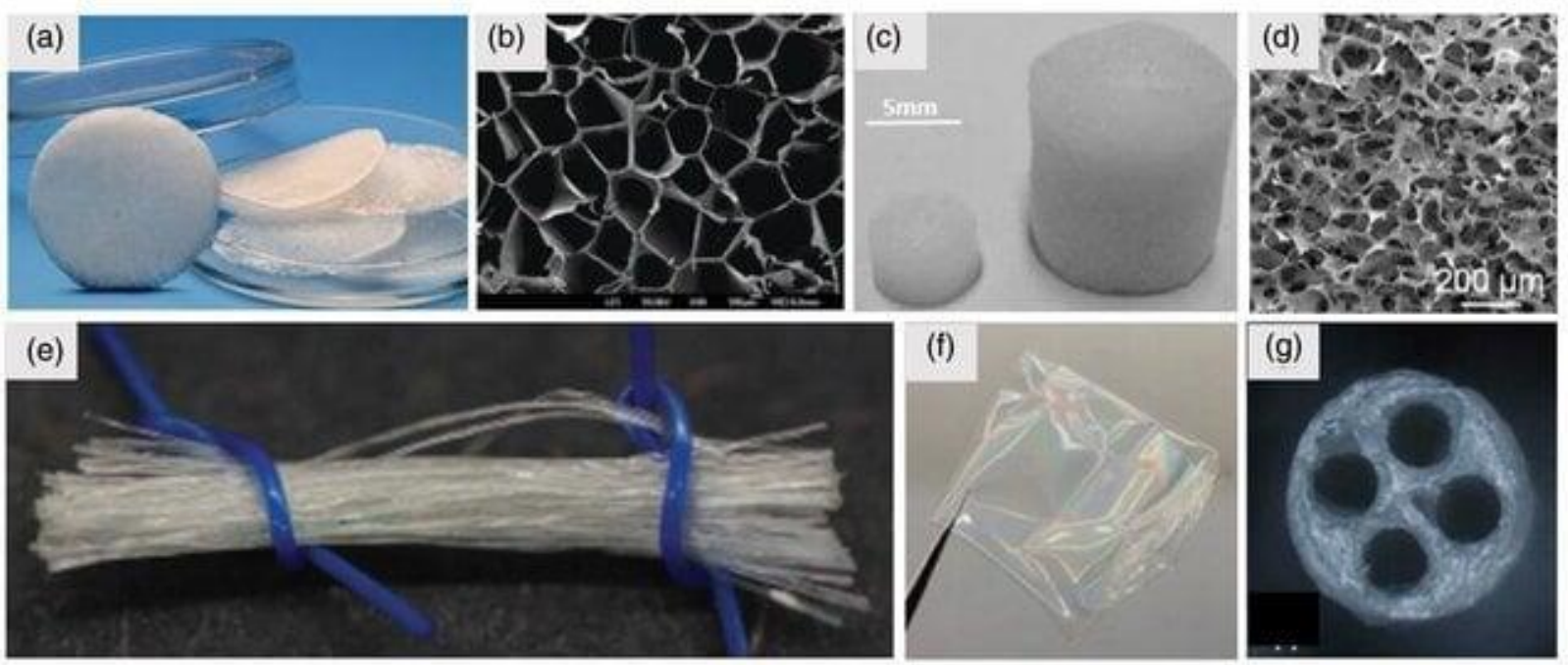


Why Collagen Is Useful in Biomaterials

- Collagen is **widely** used in **biomedical applications** because it can **mimic many features** of the natural **extracellular matrix**.
- **Useful properties**
  - Biocompatible
  - Non-toxic
  - Usually causes minimal immune response
  - Can be resorbed by the body
  - Supports cell attachment and tissue repair
  - Can be processed into different forms

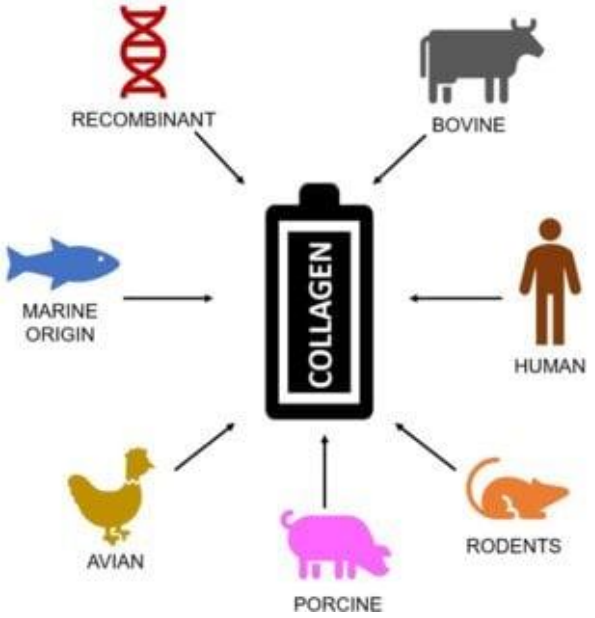
Collagen common forms

- Common forms
  - Gels
  - Porous sponges
  - Scaffolds
  - Sheets
  - Membranes



Sources of Medical Collagen

- Collagen used in medical applications can come from different biological sources.
- **Common sources**
  - Bovine sources
  - Porcine sources
  - Equine sources
  - Human donor tissues
- **Important safety point**

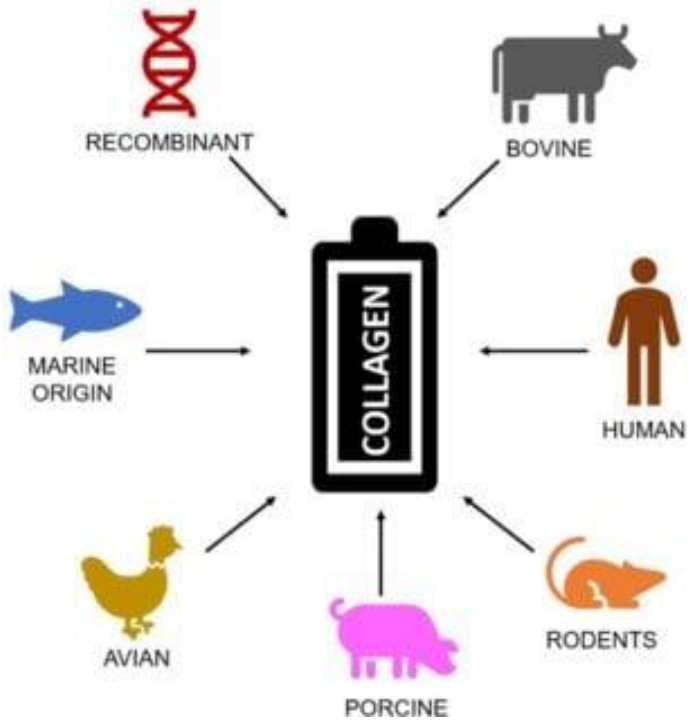


- Animal-derived collagen must be carefully processed and tested to reduce the risk of disease transmission and immune reaction.

Sources of Medical Collagen

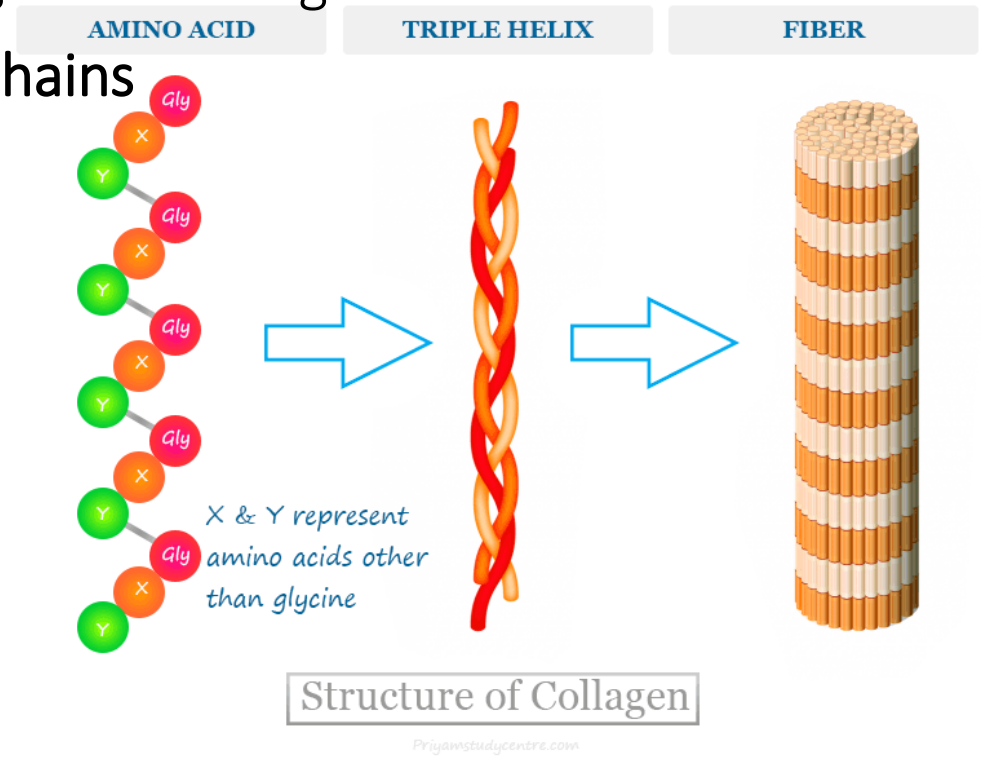
- **Animal-derived collagen** must be carefully **processed** and **tested** to reduce the **risk** of **disease transmission** and **immune reaction**.

- **Natural origin** does not automatically mean the **material** is **completely safe**. It still needs **purification**, **sterilization**, and biological **testing**.



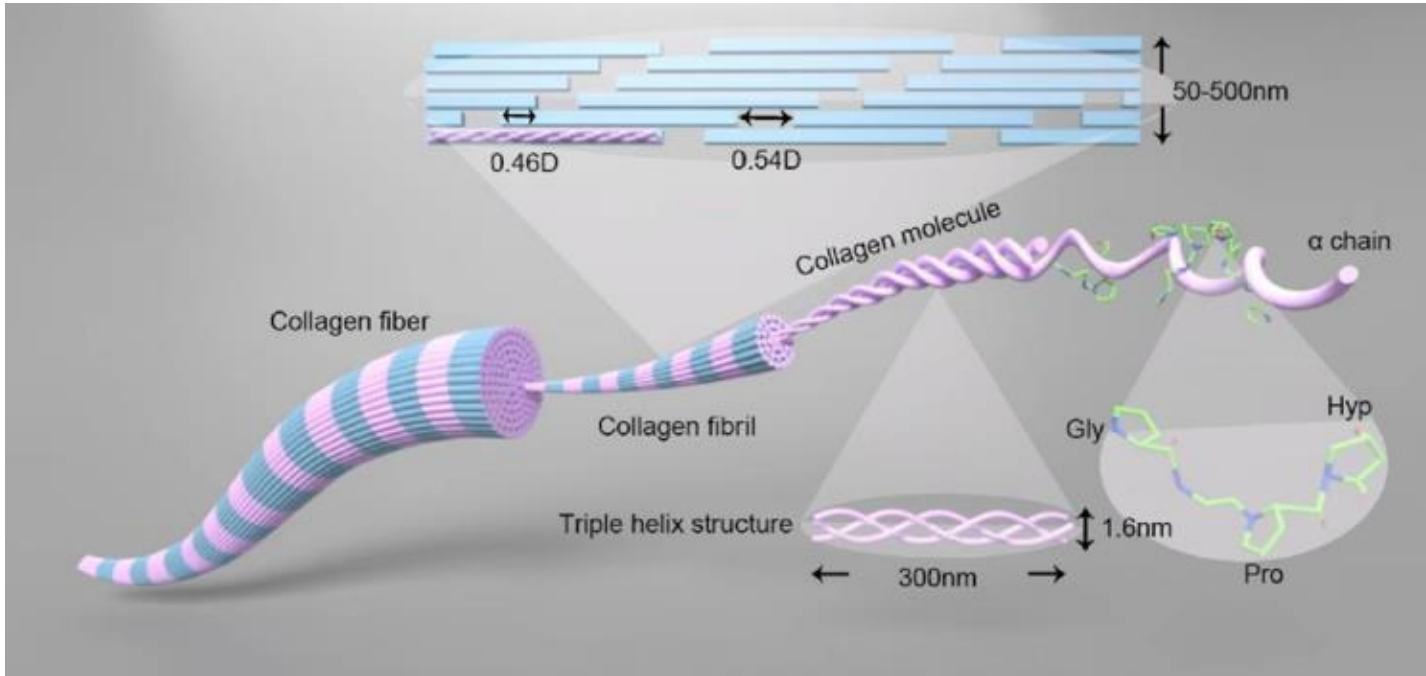
Basic Structure of Collagen

- Collagen has a special molecular structure that gives it strength.
- Collagen is made of protein chains called **alpha chains**
- Three alpha chains twist together
- This forms a **triple helix**
- The repeating amino acid sequence is usually:
  - **Glycine – X – Y**
- Where X and Y are often:
  - Proline
  - Hydroxyproline
- **Why this matters :** The triple-helix structure gives collagen high tensile strength



Hierarchical Structure of Collagen

- Collagen has a hierarchical structure, meaning small units assemble into larger structures.
- **Structure order**
  - Amino acid sequence
    → Alpha chains
    → Triple helix
    → Tropocollagen molecule
    → Collagen fibrils
    → Collagen fibers
    → Tissue structure



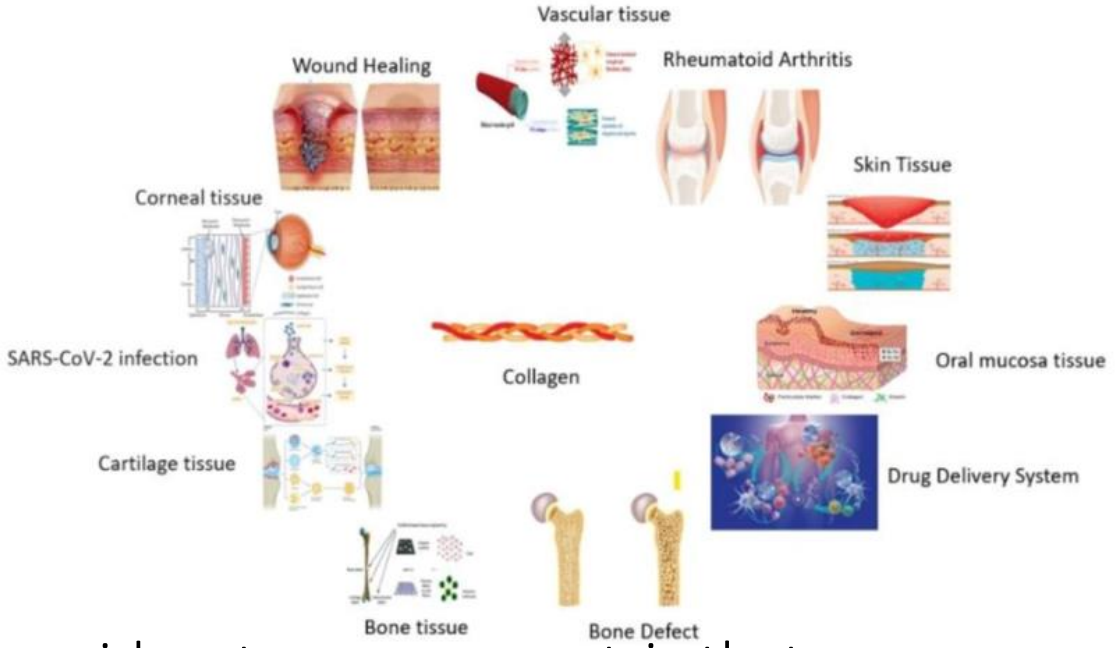
Major Types of Collagen

- Most important type in biomaterials
- Type I collagen is the most abundant and most commonly used type in biomaterial applications.

Collagen Type	Main Location	Main Function
Type I	Skin, bone, tendon, ligament, cornea	Provides tensile strength
Type II	Articular cartilage	Supports cartilage and shock absorption
Type III	Skin, blood vessels, internal organs	Provides elasticity and flexibility
Type IV	Basement membranes	Forms networks and filtration barriers

Biomedical Applications of Collagen

- Collagen is used in many medical and tissue engineering applications.
  - Wound dressings
  - Artificial skin substitutes
  - Tissue engineering scaffolds
  - Drug delivery systems
  - Bone regeneration
  - Periodontal tissue regeneration
  - Nerve guidance tubes
  - Vascular prostheses



- **Example :** In wound healing, collagen can provide a temporary matrix that supports cell migration and tissue repair.

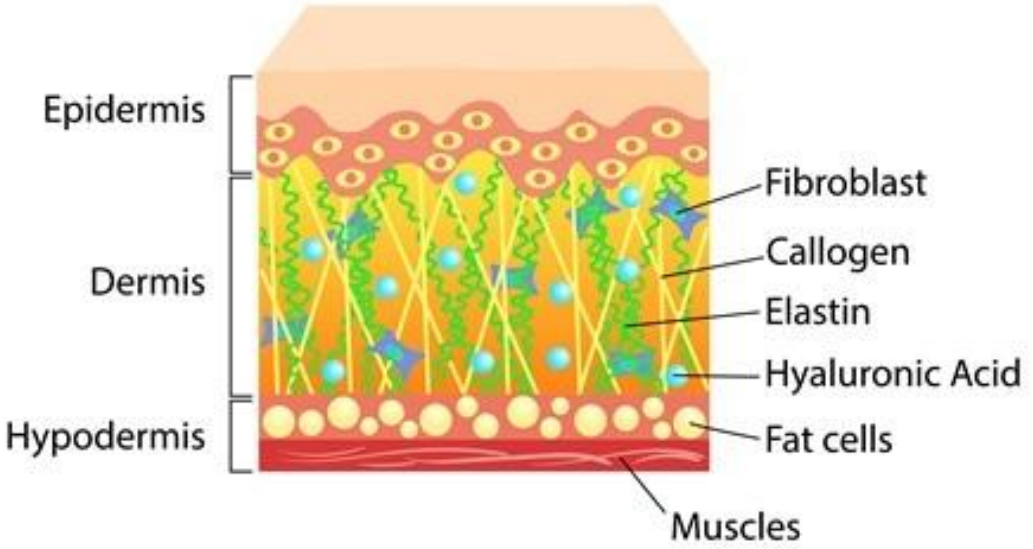
Limitations of Collagen Biomaterials

- Weak mechanical strength in some applications
- Can deform or contract when cells pull on the collagen fibers
- Degradation rate can be difficult to control
- Possible immune reaction depending on source and processing
- Batch-to-batch variation because it is naturally derived

2.Elastin

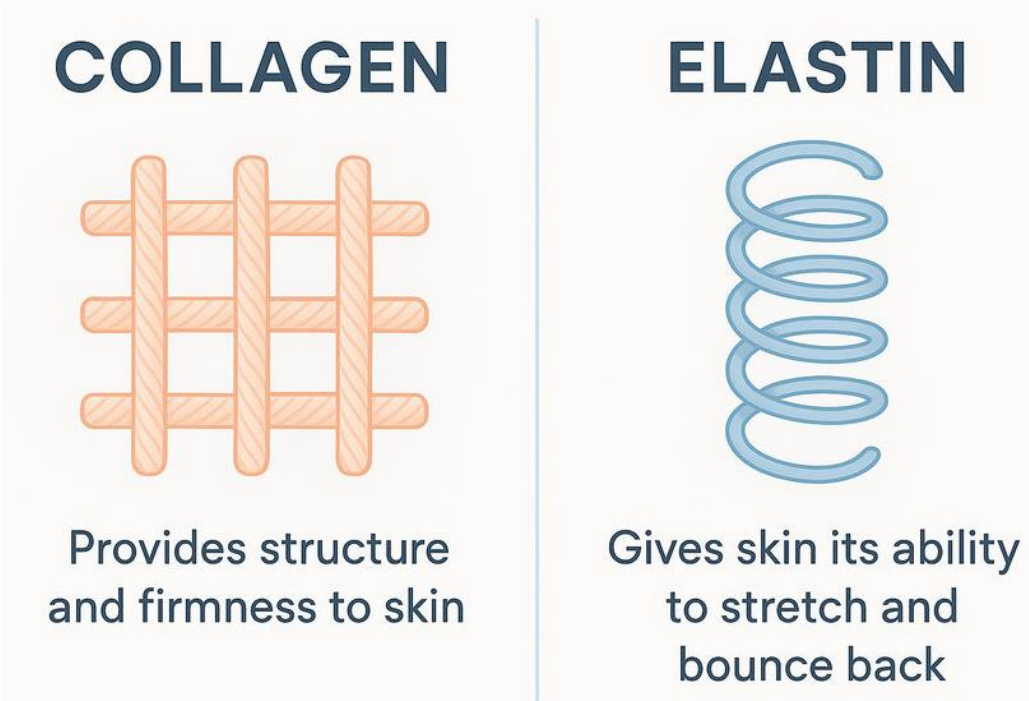
Elastin

- Elastin is a natural **protein-based** biomaterial found in the **extracellular matrix**.
- It is mainly responsible for giving tissues the ability to stretch and return to their original shape.
- Elastin is very important in tissues that undergo repeated deformation.
- Examples
  - Skin
  - Lungs
  - Arteries
  - Bladder
  - Intervertebral discs



Elastin in the Human Body

- Elastin is found in tissues where **elasticity**, **flexibility**, and **strength** are needed.
- **Approximate elastin content:**
  - Elastic ligaments: about 70%
  - Large arteries: about 50%
  - Lung tissue: about 30%
  - Skin: about 2–4%



- The amount of elastin depends on how much **elastic movement** the tissue needs.

Why Elastin Is Important

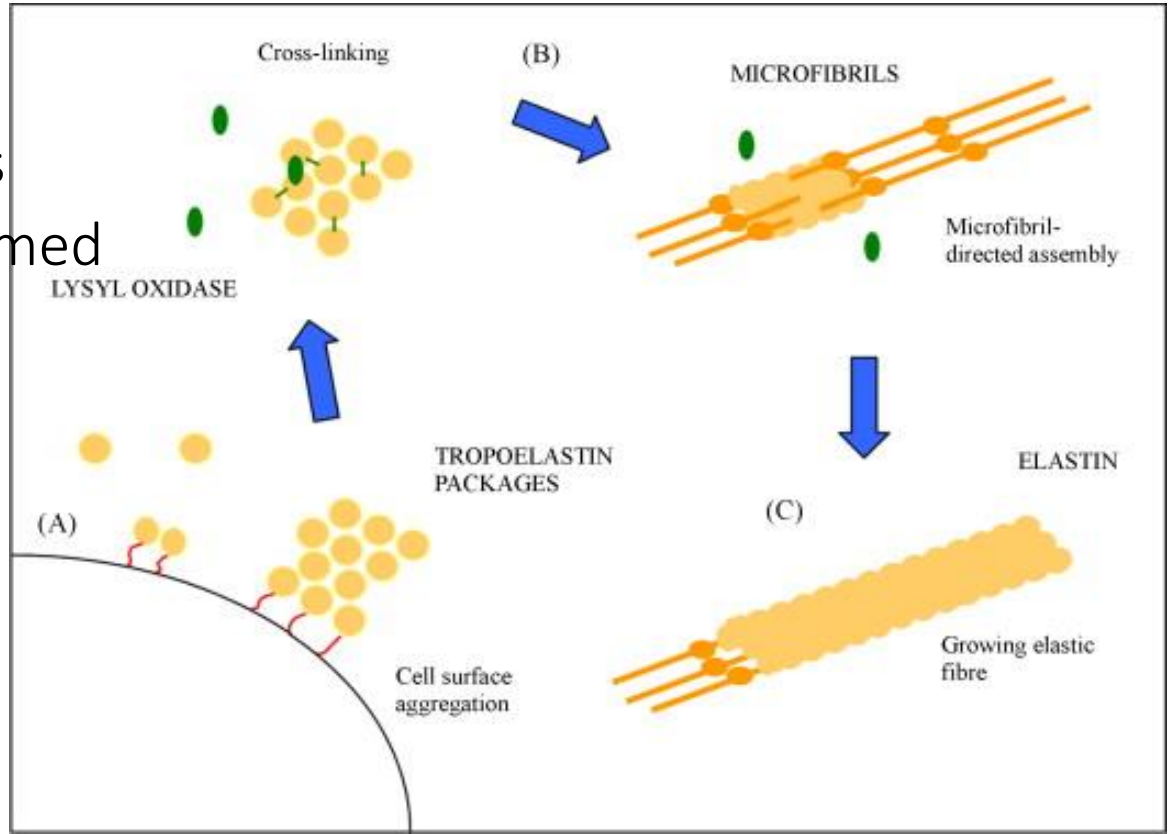
- Elastin allows tissues to:
  - Stretch without rupture
  - Return to their original shape
  - Store mechanical energy during deformation
  - Recover after stress is removed
  - Withstand repeated strain
- ***Example:*** Large arteries expand when blood is pumped from the heart and recoil again after each pulse.

Structure of Elastin

- Elastin has a different structure from collagen.
- Collagen
  - Closely packed triple-helix structure
  - Provides high tensile strength
- Elastin
  - Looser, irregular protein chains
  - Covalently cross-linked
  - Forms a rubber-like elastic network
- Key idea
  This loose cross-linked network gives elastin its elastic behavior.

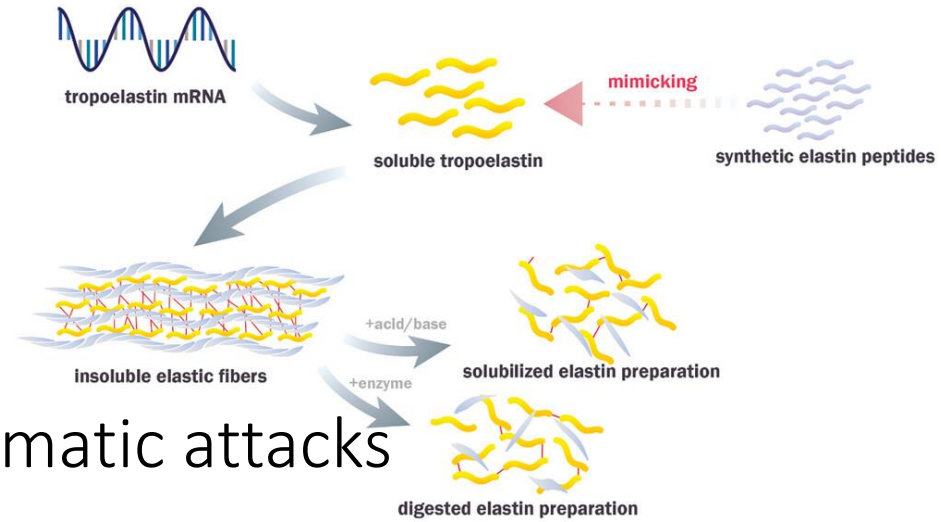
Tropoelastin and Cross-Linking

- Elastin is first produced as a soluble precursor called tropoelastin.
- **Main steps:**
  - Cells produce tropoelastin
  - Tropoelastin self-assembles
  - Lysyl oxidase enzymes create cross-links
  - A stable insoluble elastin network is formed
- **Cells that produce tropoelastin:**
  - Smooth muscle cells
  - Endothelial cells
  - Fibroblasts
  - Chondrocytes



Useful Properties of Elastin Biomaterials

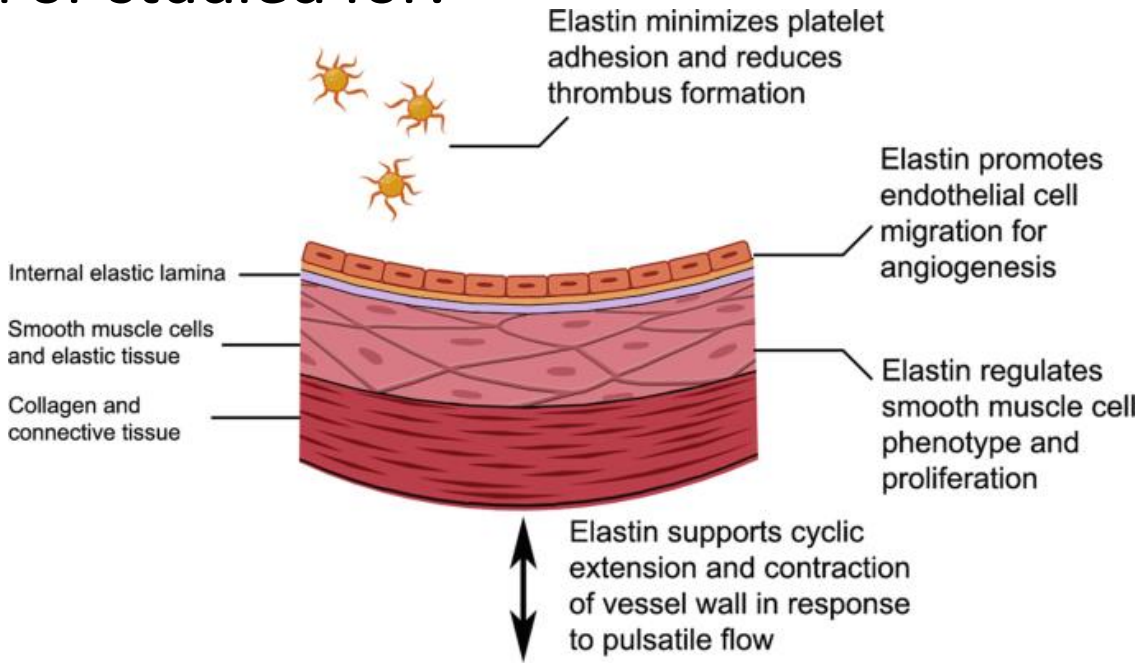
- Elastin is useful in biomaterials because it has:
  - High elasticity
  - Ability to stretch and relax
  - Good mechanical stability
  - Long biological half-life
  - Resistance to many chemical and enzymatic attacks
  - Ability to support elastic tissue behavior
- Elastin is degraded mainly by elastase.



specialized group of enzymes

Biomedical Applications of Elastin

- Elastin-based biomaterials are used or studied for:
  - Vascular stents
  - Blood vessel repair
  - Heart valve applications
  - Skin and wound repair
  - Bladder and intestinal repair
  - Lung and soft tissue engineering
  - Cosmetic implants



- Example
  In vascular applications, elastin helps the material behave more like natural arteries.

Limitations of Elastin Biomaterials

- **Main limitations:**
  - Difficult to purify from natural tissues
  - Possible contamination with other proteins
  - Calcification after implantation
  - Difficult control of degradation
  - Processing can be challenging

- **Calcification problem :** Elastin may act as a nucleation site for mineral deposition inside the body.

Synthetic Elastin-Like Polypeptides

- Because natural elastin is difficult to purify, synthetic elastin-like polypeptides can be produced.
- **They are based on repeating amino acid sequences:**
  - Valine – Proline – Glycine – X – Glycine
  - X can be changed to control material behavior.
- **Advantages:**
- Precise molecular design
- Controlled molecular weight
- Low variability
- Can be designed for drug delivery, cell adhesion, and tissue engineering

Temperature-Responsive Elastin Materials

- Some elastin-like polypeptides respond to temperature.
- At low temperature
  - They are soluble in water.
- Above a critical temperature
  - They become insoluble and aggregate.
- This temperature is called **the inverse transition** temperature.
- *Why this matters:*
  - These materials can be designed to release drugs or change shape in response to body conditions.

3.Silk

Silk



- Silk is a natural **protein-based biomaterial** produced as fibers by some organisms such as **silkworms** and **spiders**.
- It is a **structural protein** used in both **medical** and **non-medical** applications.
- Silk has long been used as a biomaterial for **sutures**.
- It is known for its **biocompatibility**, **slow degradability**, and **excellent mechanical properties**.
- Silk is another important natural biomaterial, similar to collagen and elastin, but with excellent fiber strength and long medical use.

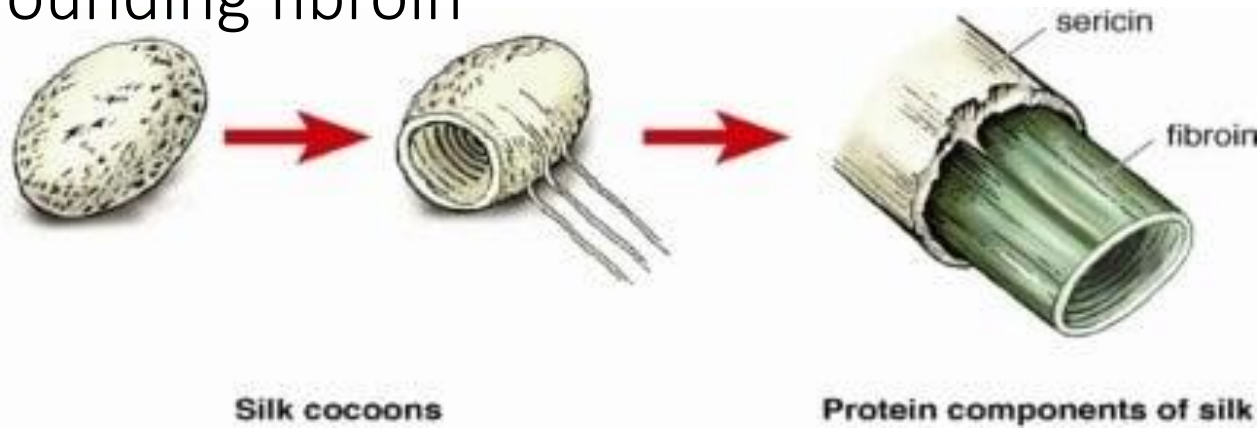
Main Source of Medical Silk

- The main medical silk is obtained from the **silkworm Bombyx mori**.
- The silkworm produces silk to form its **cocoon**.
- Raw silk fibers from the cocoon are used as a starting material for many products.
- **Bombyx mori silk** is the most commonly used silk in biomaterials.
- Although many animals produce silk, the most important source in biomedical applications is the **domesticated silkworm**.



Main Components of Silk

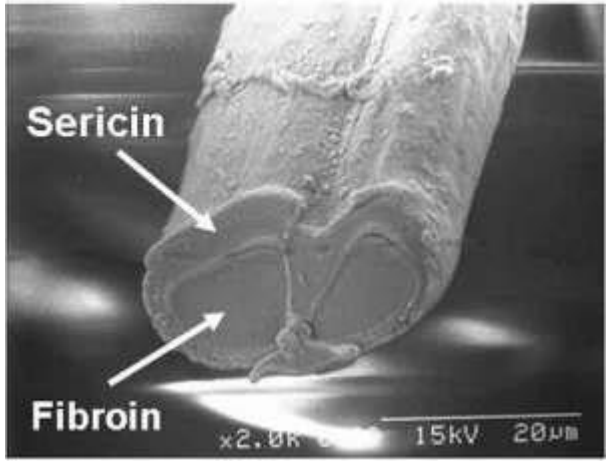
- Silk fiber is mainly composed of two proteins:
  - **Fibroin** → the main fibrous structural protein
  - **Sericin** → a glue-like protein surrounding fibroin
- In general, silk contains about:
  - **75% fibroin**
  - **25% sericin**



- The outer layers usually contain **more sericin**.

Sericin and Fibroin

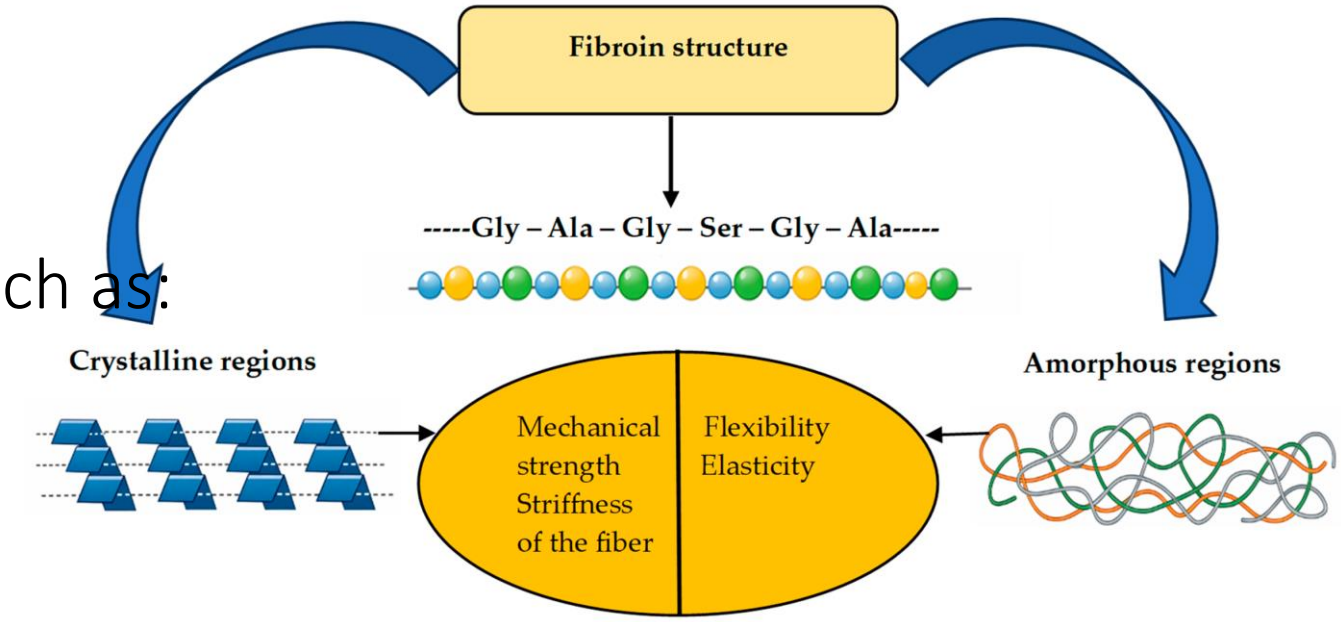
- Sericin
  - Acts as a **gum binder** that holds the silk structure together
  - More **water soluble** than fibroin
  - Has properties such as **moisture absorption**, **UV resistance**, and **antimicrobial behavior**
- Fibroin
  - Main structural protein of silk
  - Less water soluble
  - Responsible for the **strength** and **toughness** of silk fibers
  - It is a **semi-crystalline polymer**



- sericin mainly protects and binds, while fibroin gives the fiber its mechanical performance.

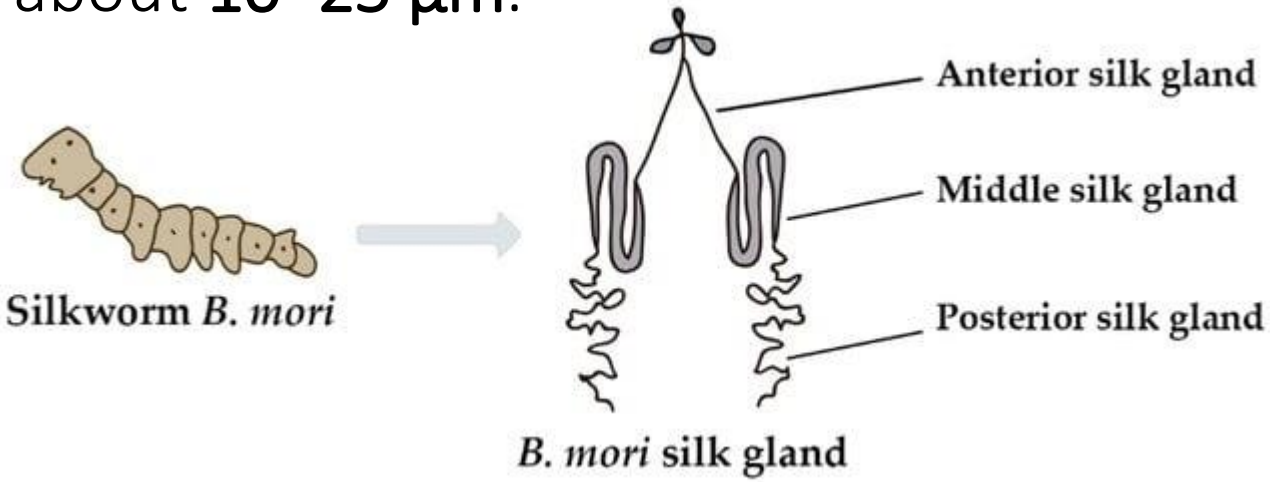
Basic Structure of Silk Fibroin

- Fibroin is made of protein chains with specific molecular organization.
- It contains:
  - Heavy chain (H-chain)
  - Light chain (L-chain)
  - P25 glycoprotein
- Fibroin is rich in amino acids such as:
  - Glycine
  - Alanine
  - Serine



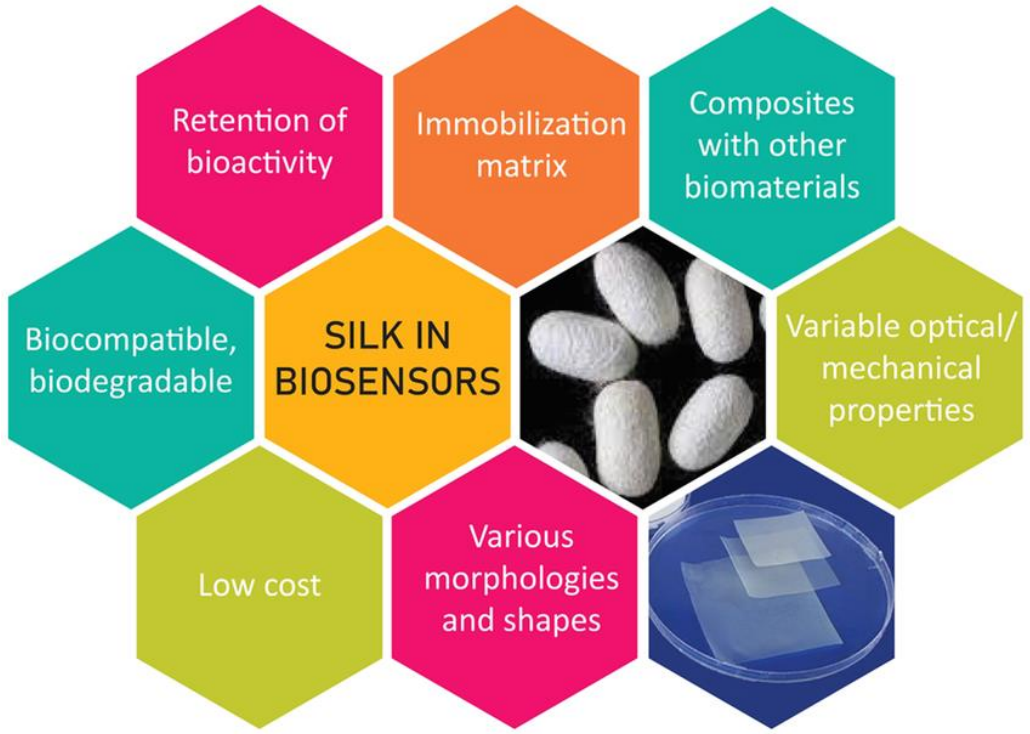
Formation of Silk Fiber

- In the silkworm gland, **fibroin** and **sericin** are produced separately.
- They mix in the **anterior section** of the gland.
- Silk is then drawn out through the **spinneret**.
- The final silk fiber is made of t**wo fibroin threads**
  - Glued together by **sericin**
  - The diameter of a silk thread is about **10–25 µm**.



Properties of Silk

- Silk has several useful properties:
  - **Biocompatible**
  - **Slowly degradable**
  - **Strong and tough**
  - **Elastic**
  - **Hygroscopic** (absorbs moisture)
  - Resistant to many **proteolytic enzymes**
  - Poor conductor of **electricity**
- Typical mechanical values for Bombyx mori silk:
  - **Tensile strength ≈ 650 MPa**
  - **Modulus ≈ 15 GPa**
  - **Elongation at break ≈ 15%–20%**



Properties of Silk

- Typical mechanical values for Bombyx mori silk:
  - **Tensile strength ≈ 650 MPa**
    - Higher than many common polymers and comparable to some engineering metals in strength
  - **Modulus ≈ 15 GPa**
    - Much less stiff than steel, but stiffer than many soft polymers
  - **Elongation at break ≈ 15%–20%**
    - Can stretch moderately before failure

Spider Silk



- Spider silk is also an important natural silk.
- It is known for being:
  - **Very strong**
  - **Highly elastic**
  - **Lightweight**
- Spider silk can have a **fracture strength of about 1100 MPa**.
- However, spider silk is difficult to **mass-produce commercially**.
- Spider silk mainly for comparison. It is impressive mechanically, but not practical for large-scale production like silkworm silk.

Biomedical Applications of Silk

- Silk is widely used in **medical applications. The most common use is sutures.**
- Silk sutures are generally used for:
  - **Soft tissue approximation**
  - **Ligation**
- Applications include:
  - **Cardiovascular surgery**
  - **Ophthalmic surgery**
  - **Neurological surgery**
- Other biomedical applications include:
  - **Drug delivery**
  - **Tissue engineering**



Why Silk Is Useful in Biomaterials

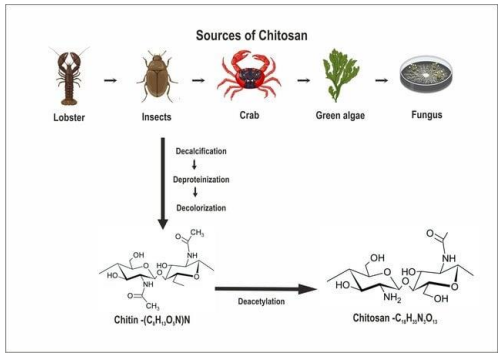
- Silk is useful in biomaterials because it has:
  - Good **biocompatibility**
  - **Slow degradation**
  - Excellent **mechanical strength**
  - Good **elasticity**
  - Ability to be processed into different forms
  - Silk can support biomedical uses where both **strength** and **biological compatibility** are needed.

4.Chitosan

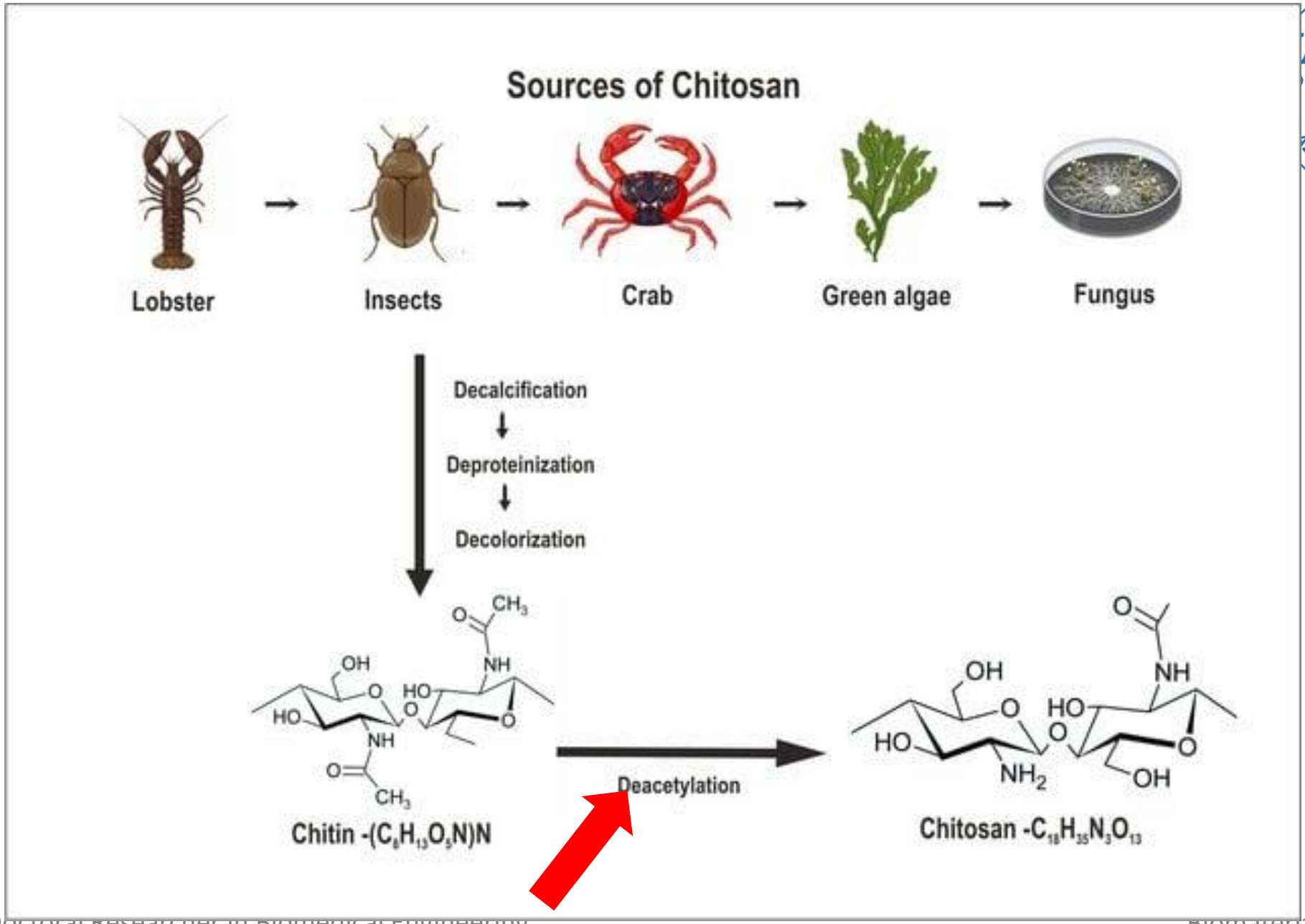
Chitosan



- Chitosan is a natural **polysaccharide-based biomaterial**.
- It is derived from **chitin**, which is found in:
  - Cell walls of some **fungi**
  - Cuticles of **insects**
  - **Exoskeletons** of shellfish such as **shrimp and crabs**



- **Chitosan** is important because it is **biocompatible**, **biodegradable**, and easily processed into different forms.
- **Chitosan** as a **natural polymer**, but unlike **collagen** and **silk**, it is not **protein-based**. It is **polysaccharide-based**.


Sources of Chitosan
Lobster
Insects
Crab
Green algae
Fungus
Decalcification
Deproteinization
Decolorization
Chitin -(C8H13O5N)N
Deacetylation
Chitosan -C18H35N3O13


Crustacean shells
Chitin
Chitosan

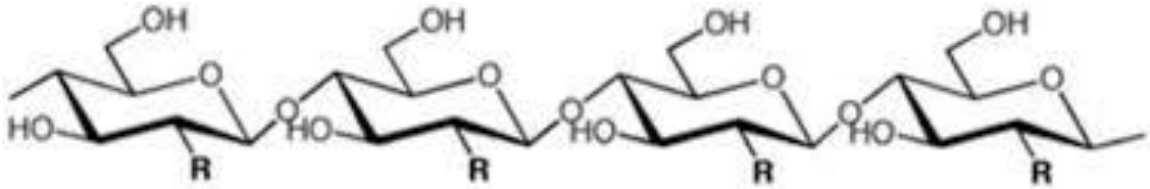
Chitin and Chitosan



- **Chitin** is the parent material of chitosan.
- Chitin is a long linear polymer made mainly of **N-acetyl-D-glucosamine**.
- Chitosan is produced by **deacetylation** of chitin.
- **Deacetylation** means removing the **acetyl group** from chitin.
- When more than **50%** of the acetyl groups are removed, the material is generally called **chitosan**.
- **Chitosan** is **chemically** modified **chitin**. The **removal** of **acetyl groups** changes its properties.

Basic Structure of Chitosan

- **Chitosan** is a linear **polysaccharide**.
- It contains two types of repeating units
  - **D-glucosamine**
  - **N-acetyl-D-glucosamine**



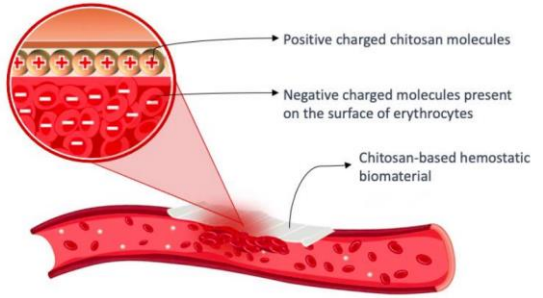
Cellulose: **R** = OH
Chitin: **R** = $NHCOCH_3$
Chitosan: **R** = NH_2

- These units are connected by **β-1,4 glycosidic bonds**.
- The structure of chitosan strongly affects its solubility, degradation, and biological behavior.

Degree of Deacetylation

- Chitosan is produced by removing acetyl groups from chitin.
- **DDA tells us how much deacetylation happened.**
- Higher DDA means:
  - More amino groups
  - More positive charge in acidic conditions
  - Different solubility and degradation behavior
  - Different interaction with cells and tissues
- So; **DDA** is like a “control parameter” for chitosan. It changes how chitosan behaves as a biomaterial.

Why Chitosan Is Positively Charged?



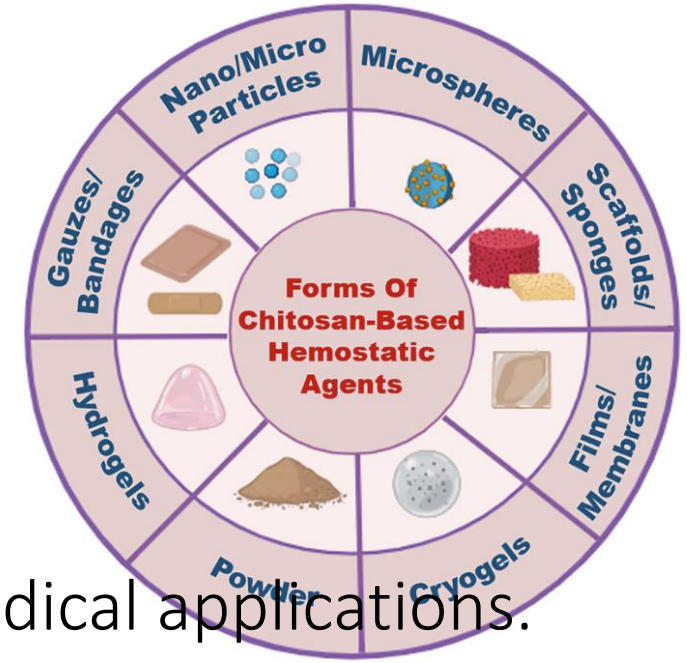
- Chitosan contains many **amino groups**.
- At acidic or near-neutral pH, these amino groups can become **protonated**.
- This makes chitosan a **positively charged polymer**.
- Because of this positive charge, chitosan can interact with negatively charged biological surfaces.
- This is important for **bioadhesion** and **hemostatic behavior**.
- This is one of the most important concepts. Chitosan's positive charge explains many of its biomedical uses, especially wound dressings.

Useful Properties of Chitosan

- Chitosan has several favorable properties for biomaterials:
  - Biocompatible
  - Biodegradable
  - Minimal foreign body reaction
  - Antimicrobial behavior
  - Supports cell adhesion and migration
  - Can be chemically modified
  - Can be processed into many forms
- **chitosan** is **attractive** in **medical applications**. It is not only **biodegradable** but also **easy to modify**.

Processing Forms of Chitosan

- Chitosan can be fabricated into different forms, such as:
  - Films
  - Fibers
  - Gels
  - Sponges
  - Microspheres
  - Nanofibers
  - Three-dimensional porous scaffolds



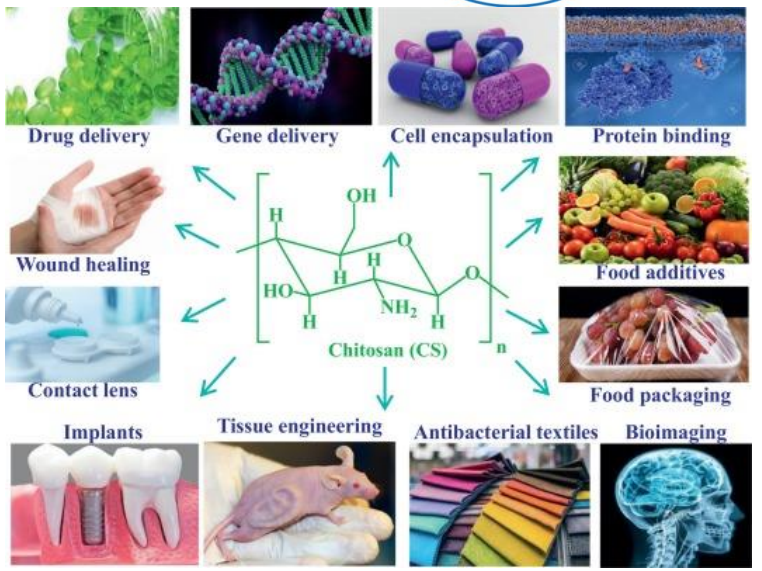
- This makes chitosan useful for different biomedical applications.
- processing flexibility is one of the major advantages of chitosan as a biomaterial.

Degradation of Chitosan

- **Chitosan can degrade in the body by:**
  - Lysozyme activity
  - Hydrolysis
  - Degradation products are simple sugars.
- The degradation rate depends mainly on:
  - Degree of deacetylation
  - Crystallinity
  - Molecular weight
  - Sterilization method
- Higher deacetylation usually increases crystallinity and slows degradation.

Biomedical Applications of Chitosan

- Chitosan is used in several biomedical applications:
  - Wound dressings
  - Hemostatic materials to stop bleeding
  - Drug delivery systems
  - Tissue engineering scaffolds
  - Cartilage repair materials



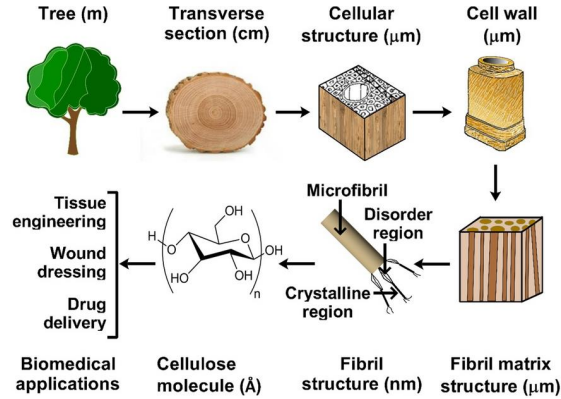
- Examples include chitosan-based hemostatic dressings and cartilage repair gels.
- The most important applications to emphasize are wound dressing, bleeding control, drug delivery, and tissue engineering.

Limitations of Chitosan

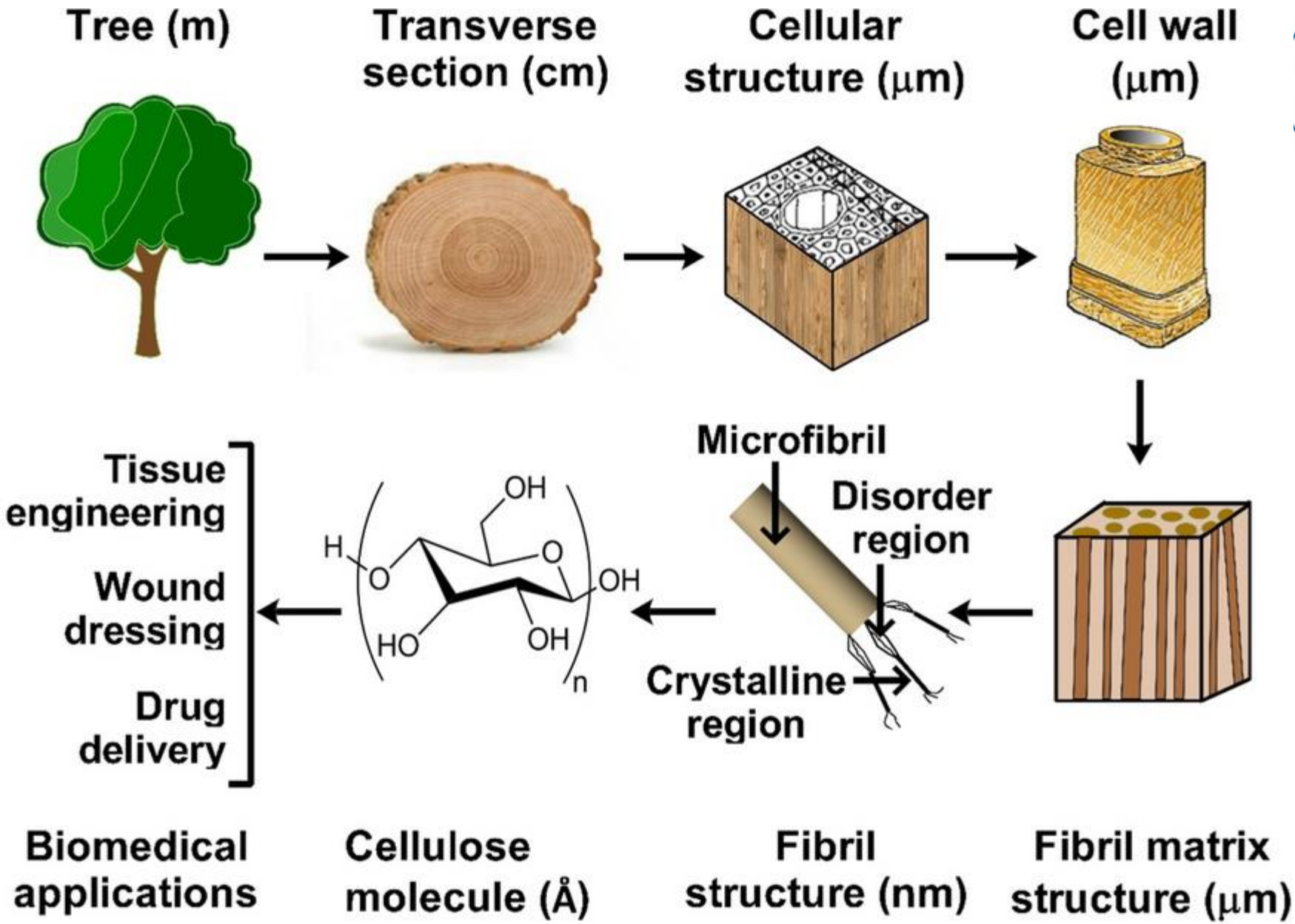
- Although chitosan is useful, it also has limitations.
  - Mechanically weak compared with metals and ceramics
  - Can swell in physiological solutions
  - May show inconsistent cell response
  - Properties depend strongly on source and processing
  - Degradation and strength may be difficult to control

5.Cellulose

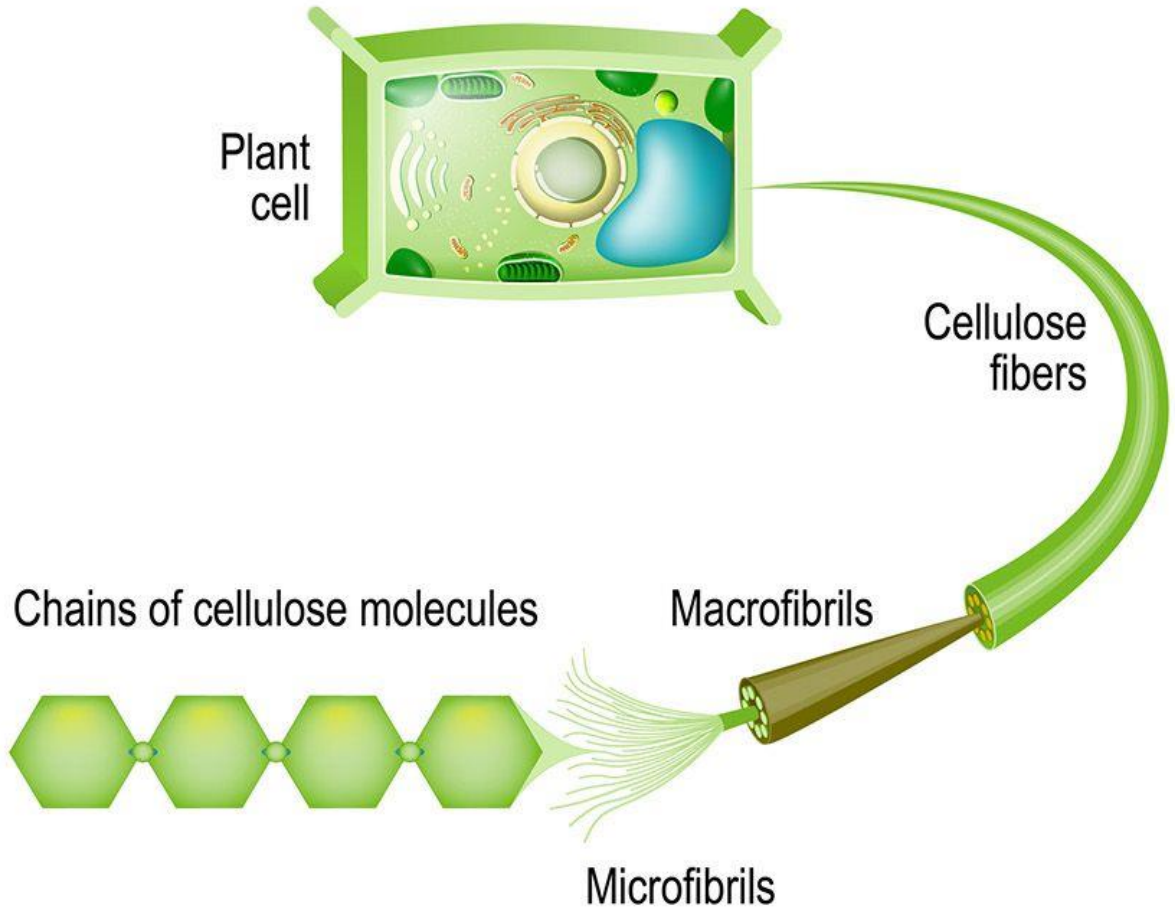
Cellulose



- Cellulose is a natural **polysaccharide-based biomaterial**.
- Its general formula is **$(C_6H_{10}O_5)n$**.
- It is a **linear polymer** made of many glucose units.
- Cellulose is the most abundant **organic polymer** in the world.
- It is mainly found in the **cell walls of green plants**.
- **cellulose** as another **natural polymer**. **Unlike collagen** and **silk**, it is not **protein-based;** it is a **polysaccharide**.

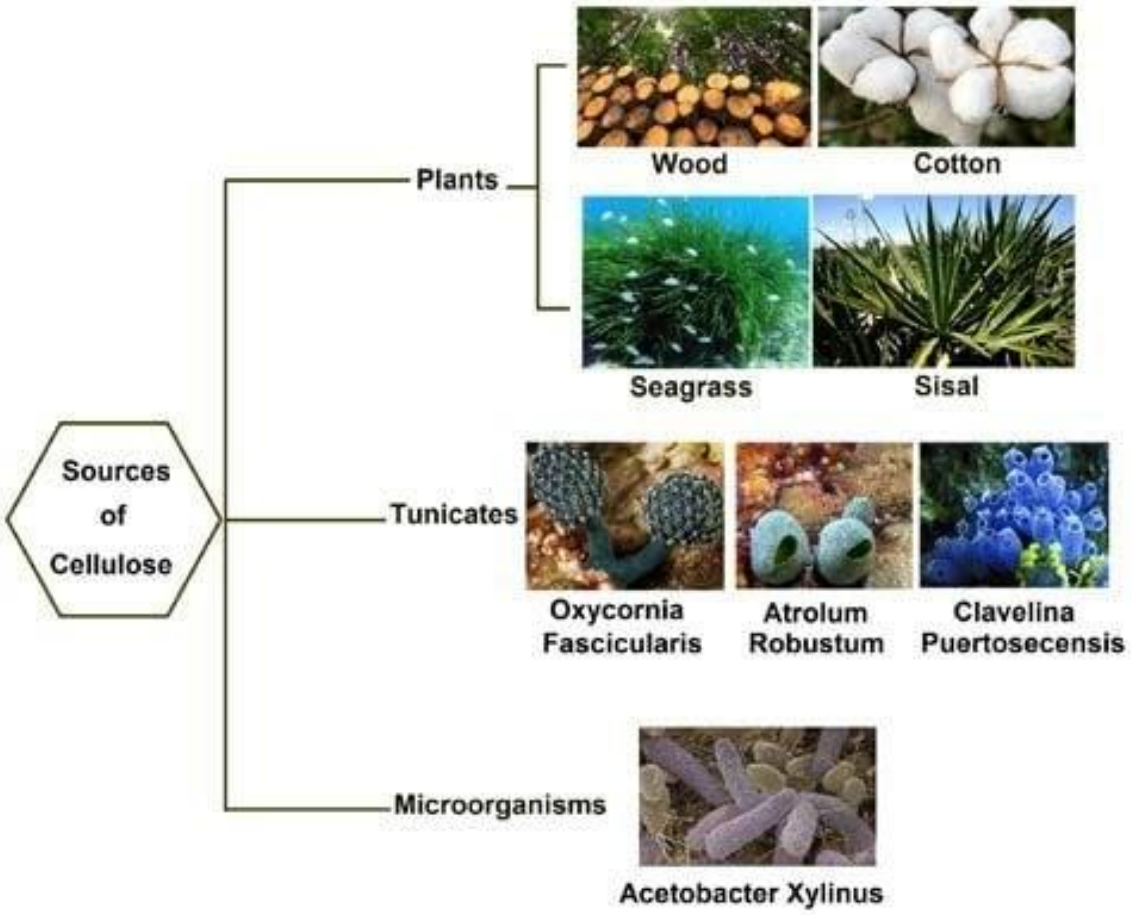

Tree (m)
Transverse section (cm)
Cellular structure (μm)
Cell wall (μm)
Microfibril
Disorder region
Crystalline region
Tissue engineering
Wound dressing
Drug delivery
OH
H
O
HO
n
Biomedical applications
Cellulose molecule (Å)
Fibril structure (nm)
Fibril matrix structure (μm)

CELLULOSE


Plant cell
Cellulose fibers
Chains of cellulose molecules
Macrofibrils
Microfibrils

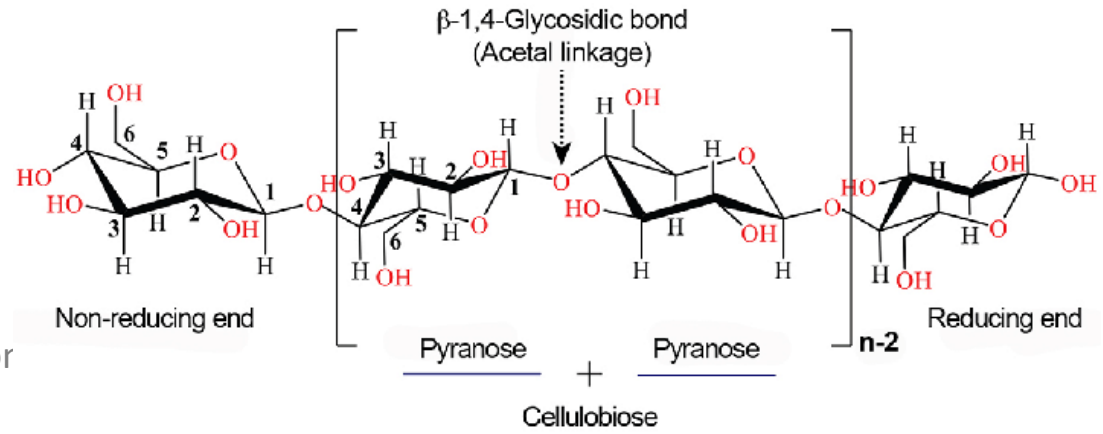
Natural Sources of Cellulose

- Cellulose is found in almost all **plant materials**.
- It is the main structural component of **plant cell walls**.
- Cellulose can also be produced by:
  - Some bacteria
  - Some algae
  - Some prokaryotes
- Bacterial cellulose is also important in biomedical applications.



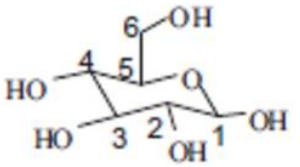
Basic Structure of Cellulose

- Cellulose is made of repeated **D-glucose units**.
- The glucose units are connected by **β-1,4 glycosidic bonds**.
- Each glucose unit is linked from **C-1** of one glucose to **C-4** of the next glucose.
- This gives cellulose a long, linear chain structure.
- The repeating units are called **anhydroglucose units**.

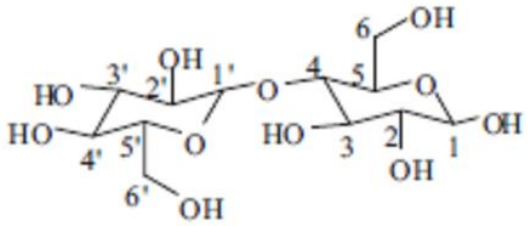


Cellulose and Cellobiose

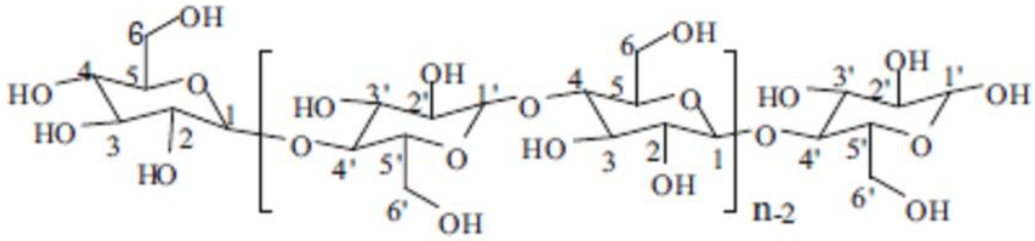
- In **cellulose**, every other glucose unit is rotated by about **180°**.
- This **arrangement** forms repeating disaccharide units called **cellobiose**.
- Cellobiose is the repeating structural unit of cellulose.
- The orientation of the glucose rings makes cellulose chains **rigid and rod-like**.



Glucose



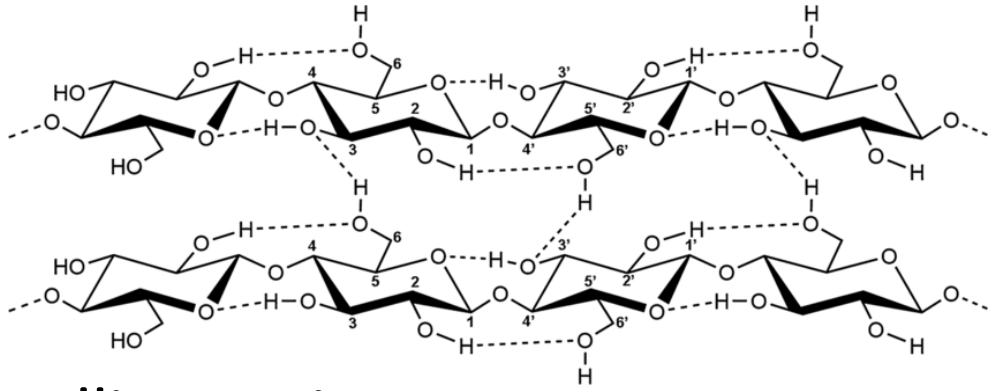
Cellobiose



Cellulose

Hydrogen Bonding in Cellulose

- Cellulose contains many **hydroxyl groups (-OH)**.
- These groups form strong **hydrogen bonds**.
- Hydrogen bonding occurs:
  - Within the same cellulose chain
  - Between neighboring cellulose chains



- This bonding helps form highly ordered **crystalline regions**.
- These crystalline regions give cellulose high **fiber strength**.
- Strong hydrogen bonding explains cellulose strength and insolubility.

Why Cellulose Is Insoluble in Water

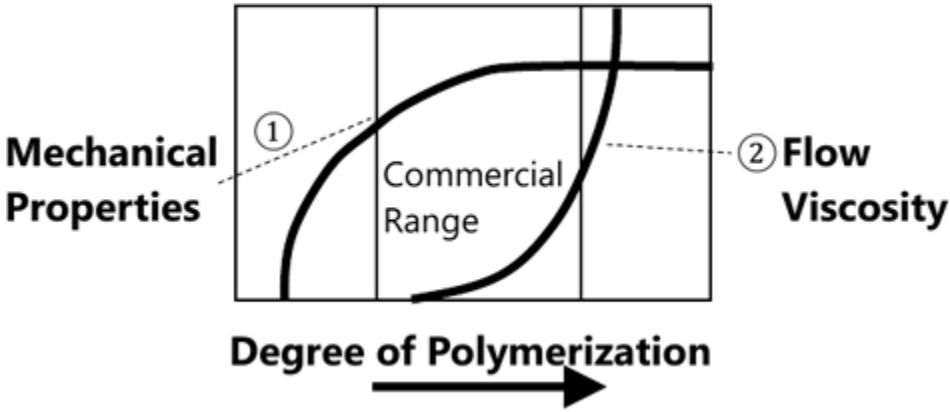
- Glucose is water soluble, but cellulose is **not water soluble**.
- This is because cellulose chains are strongly held together by **hydrogen bonding**.
- The crystalline structure prevents water from separating the chains easily.
- However, cellulose is still **hydrophilic**.
- Less crystalline regions can absorb water and cause **swelling**.

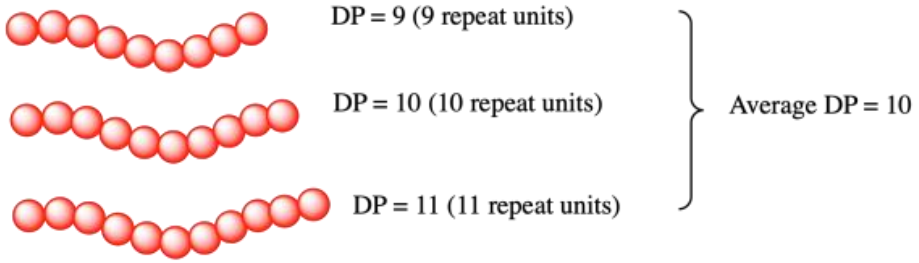
General Properties of Cellulose

- Cellulose is:
  - Low cost
  - Readily available
  - Tasteless
  - Odorless
  - Hydrophilic
  - Strong as a fiber
  - Hygroscopic, meaning it can absorb moisture
  - Its viscosity depends mainly on **molecular weight** and **degree of polymerization**.

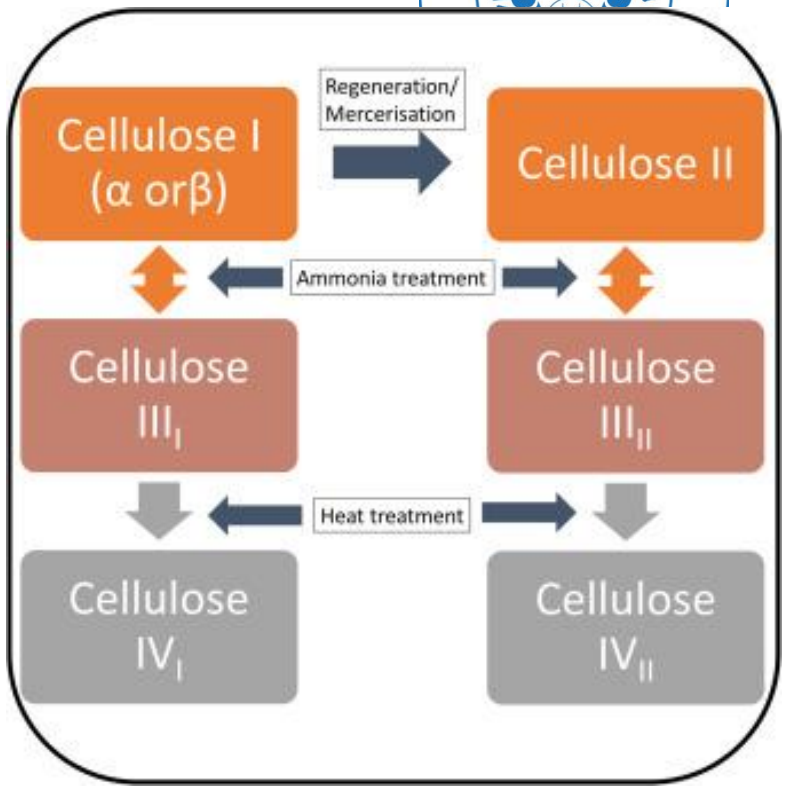
Degree of Polymerization

- The **degree of polymerization** means the number of glucose units in the cellulose chain.
- A higher degree of polymerization gives:
  - Higher molecular weight
  - Higher viscosity
  - Different processing behavior
- The degree of polymerization depends on:
  - Cellulose source
  - Isolation method
  - Processing conditions
- Longer chains usually mean higher molecular weight and higher viscosity.





Crystalline Forms of Cellulose



- Cellulose can exist in different crystalline forms.
- The main forms are:
  - **Cellulose I**
  - **Cellulose II**
  - **Cellulose III**
  - **Cellulose IV**
- **Cellulose I** is the natural form found in most plants.
- **Cellulose II** is often obtained after chemical treatment or regeneration.
- Different forms have different molecular arrangements and stability.

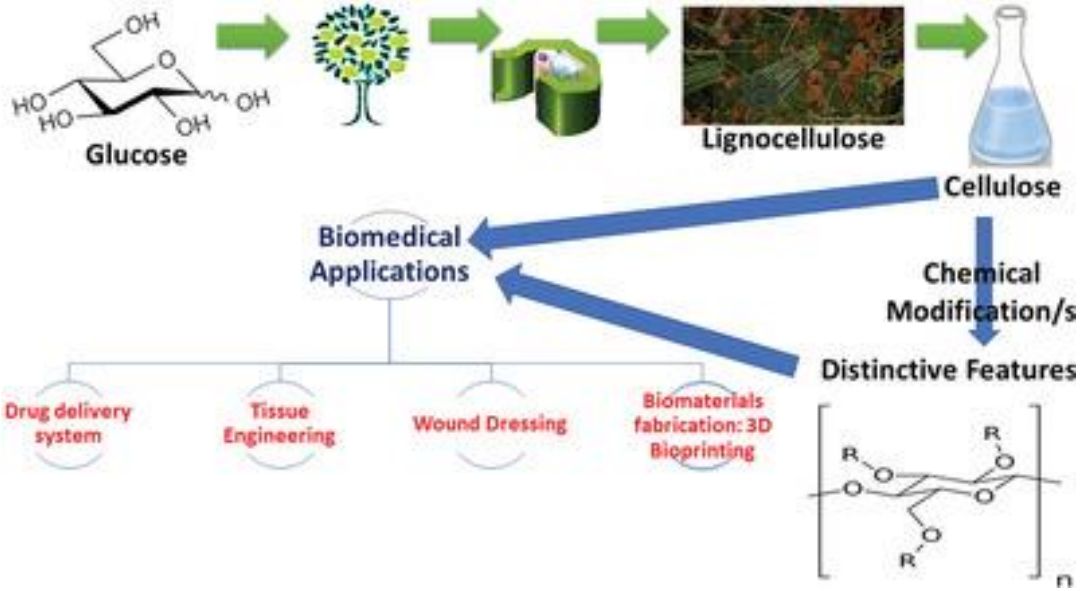
Cellulose Derivatives

- Cellulose can be chemically modified to produce useful derivatives.
- Common cellulose derivatives include:
  - Cellulose esters
  - Cellulose ethers
  - Graft copolymers
  - Cross-linked cellulose
- These derivatives can improve solubility, viscosity, and processing behavior.
- This makes cellulose useful in pharmaceutical and biomedical products.

Biomedical and Pharmaceutical Uses

- Cellulose and its derivatives are used in many medical applications.
- Common uses include:
  - Tablet binders
  - Drug delivery matrices
  - Sustained-release formulations
  - Artificial skin for burns and ulcers
  - Hemodialysis membranes
  - Hemostatic materials to control bleeding
- Ex: Low-viscosity types can be used for immediate drug release, while higher-viscosity types can be used for sustained release.

Why Cellulose Is Useful in Biomaterials



- Cellulose is useful because it is:
  - Natural and abundant
  - Low cost
  - Biocompatible in many applications
  - Strong as a fiber
  - Hydrophilic
  - Chemically modifiable
  - Available in many derivative forms
  - Suitable for films, membranes, dressings, and drug delivery systems

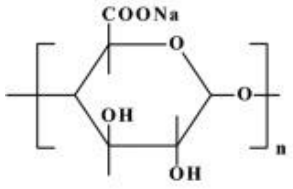
Limitations of Cellulose

Main limitations:

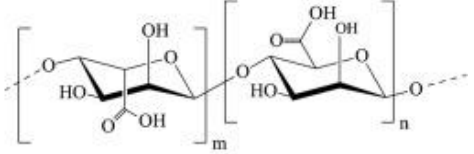
- Poor solubility in water
- Difficult processing in native form
- Properties depend on source and isolation method
- Chemical purification can degrade the cellulose chain
- Swelling can occur in less crystalline regions
- Mechanical and biological performance may require modification

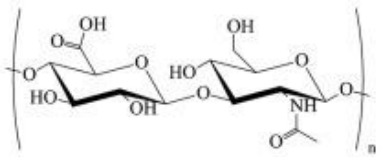
6.Alginate

Alginate



Sodium alginate





Alginic acid

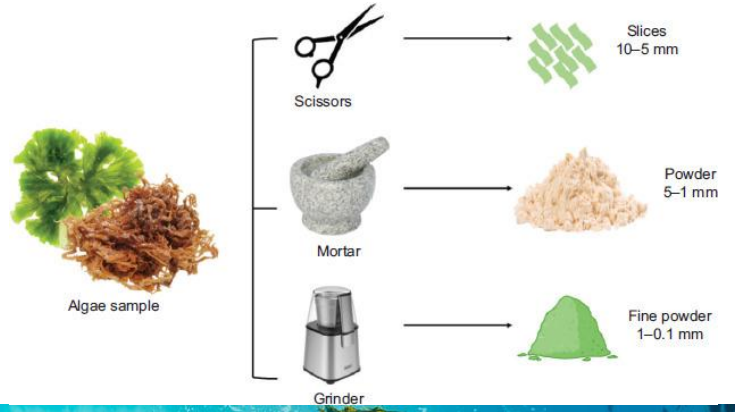


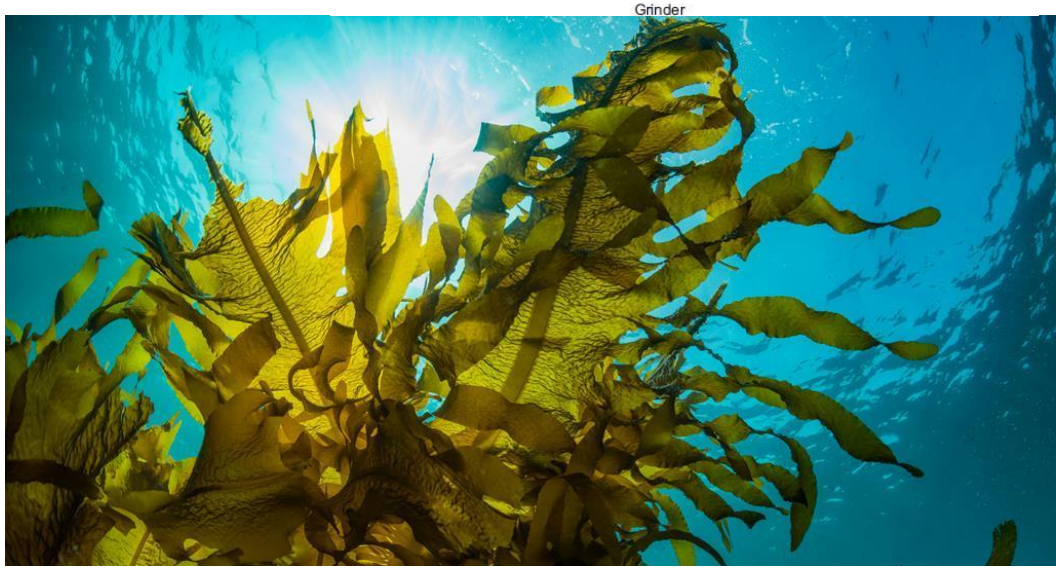


- Alginate is a natural **polysaccharide-based biomateria**
- It is a **non-branched anionic polymer**.
- It is also known as:
  - **Alginic acid**
  - **Algin**
- Alginate is important because it can form **hydrogels** easily.
- This makes it useful in **wound dressings**, **drug delivery**, and **tissue engineering**.

Natural Sources of Alginate

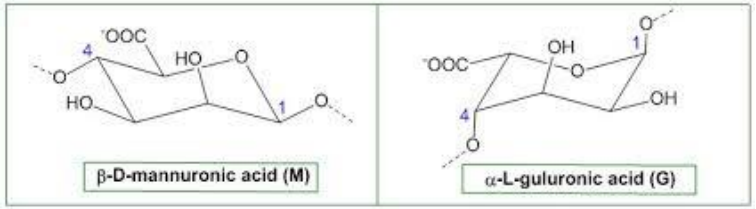
- Alginate is mainly found in **marine algae**, especially **brown seaweeds**.
- It can also be produced by some bacteria such as:
  - **Pseudomonas**
  - **Azotobacter**
- The composition of alginate depends on:
  - Source
  - Seaweed species
  - Plant age
  - Part of the seaweed used



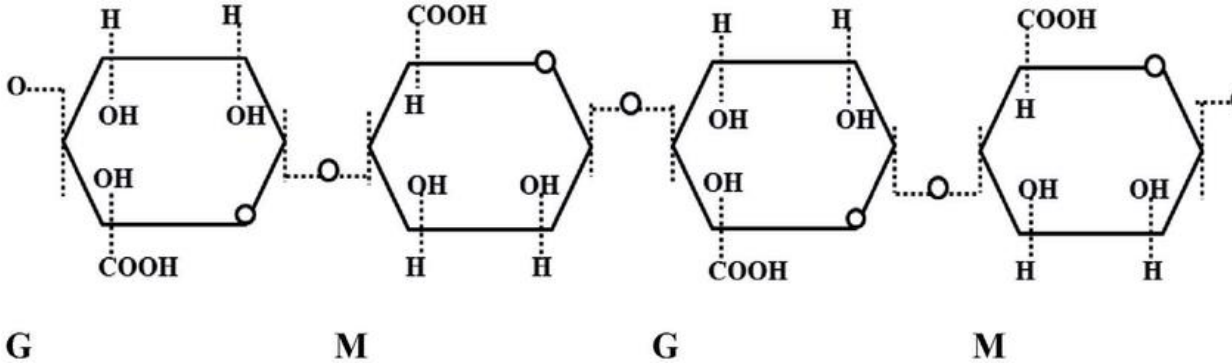


Basic Structure of Alginate

- Alginate is made of two main monomers:
  - β-D-mannuronic acid (M)
  - α-L-guluronic acid (G)

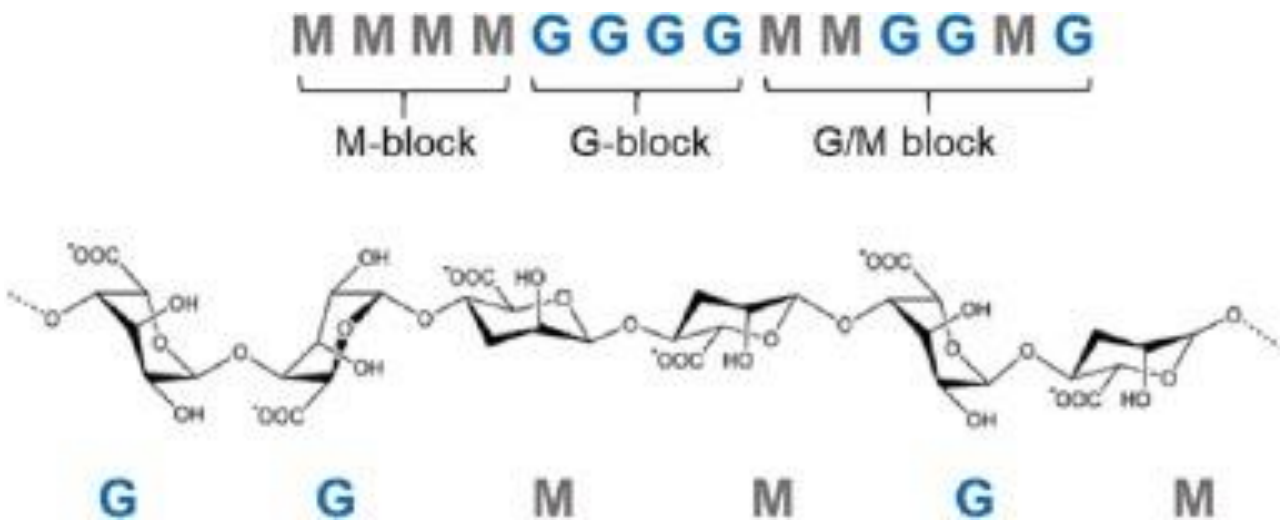


- These monomers are linked by **1–4 glycosidic bonds**.
- M and G units are arranged in blocks along the polymer chain.
- The M/G ratio strongly affects alginate properties.
- Alginate is made of M and G blocks.



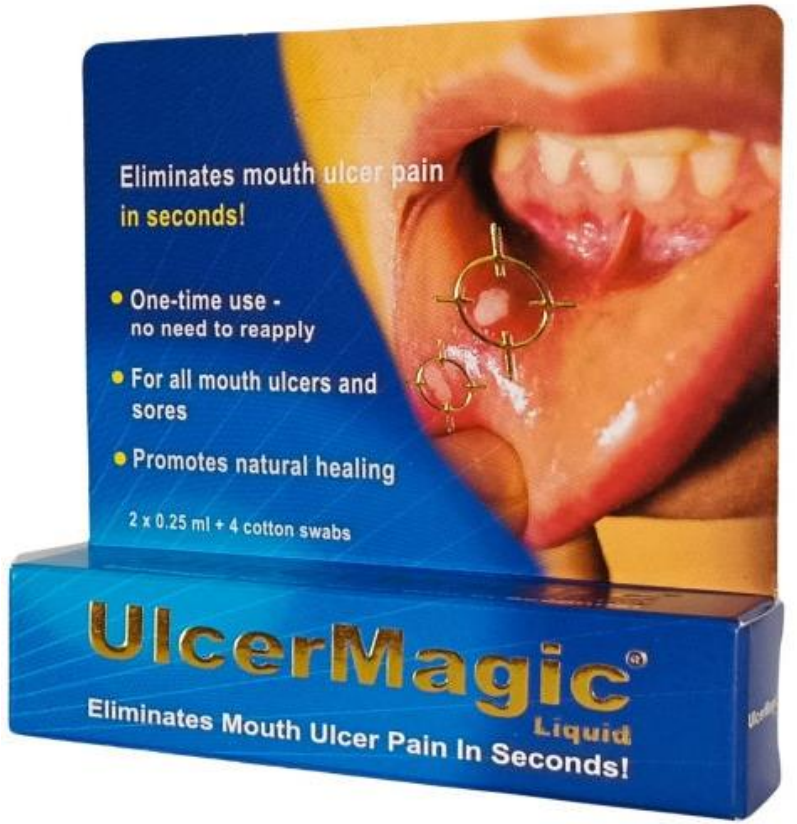
Alginate Block Structure

- Alginate can contain different sequence blocks:
  - **M-blocks**: M–M–M
  - **G-blocks**: G–G–G
  - **MG-blocks**: G–M–G–M
- These different blocks affect:
  - Gel formation
  - Mechanical strength
  - Porosity
  - Swelling behavior
  - Stability of the final gel



Useful Properties of Alginate

- Alginate has several favorable biomaterial properties:
  - Hydrophilic
  - Fairly non-toxic
  - Non-inflammatory
  - Biodegradable
  - Easily processed in water
  - Absorbs water quickly
  - Has controllable porosity
  - Can be linked to biologically active molecules
  - Acts as a good **mucoadhesive** material



A material can stick to mucosal surfaces.

Alginate Gel Formation

- Alginate can form hydrogels when cross-linked with **divalent or multivalent cations**.
- The most common ion used is **calcium ion (Ca^{2+})**.
- Calcium ions bind between **G-blocks** of neighboring alginate chains.
- This creates a **three-dimensional network**.
- The result is a water-containing **alginate hydrogel**.
- Calcium cross-linking is the basis of many alginate applications.

Effect of G and M Content

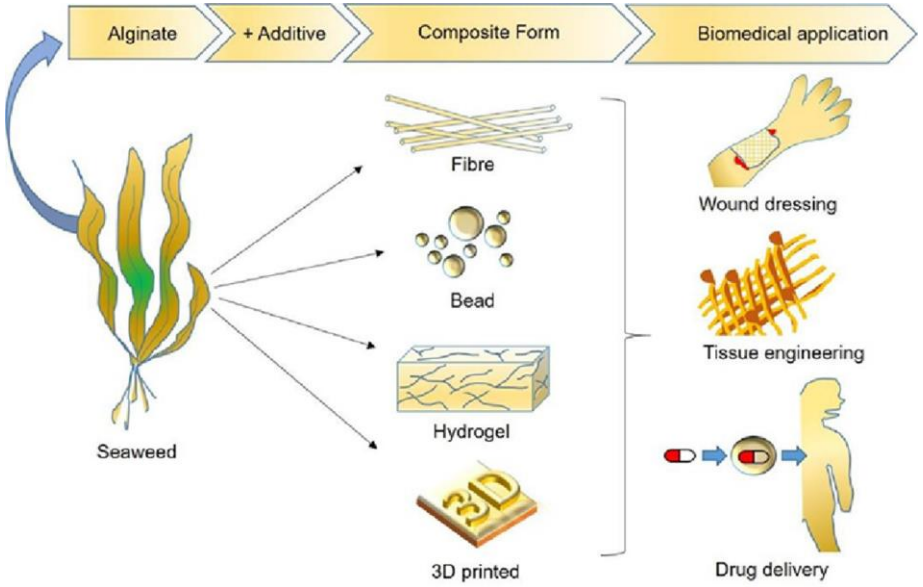
- The amount of **G** and **M** units affects the gel properties.
- High **G-content** alginate usually forms:
  - Stiffer gels
  - More porous gels
  - Better shape retention
  - Longer stability
- High **M-content** alginate usually forms:
  - Softer gels
  - Less porous gels
  - More swelling and shrinking
  - Faster loss of shape over time
- G-rich alginate is generally better when shape stability is important, such as in tissue engineering scaffolds.

Effect of Cations

- Not all ions form alginate gels in the same way.
- **Monovalent ions** do not usually cause gelation.
- Some divalent ions form stronger gels than others.
- **Ca^{2+}** is commonly used for alginate gel formation.
- **Ba^{2+}** and **Sr^{2+}** can produce stronger gels than Ca^{2+}.
- **Mg^{2+}** does not effectively form alginate gels.
- The type of ion affects the strength and formation of alginate gels.

Forms of Alginate

- Alginate can be processed into different forms:
  - Cross-linked gels
  - Pastes
  - Compounds
  - Beads
  - Fibers
  - Bandages
- These forms make alginate useful in both **medical** and **non-medical** applications.



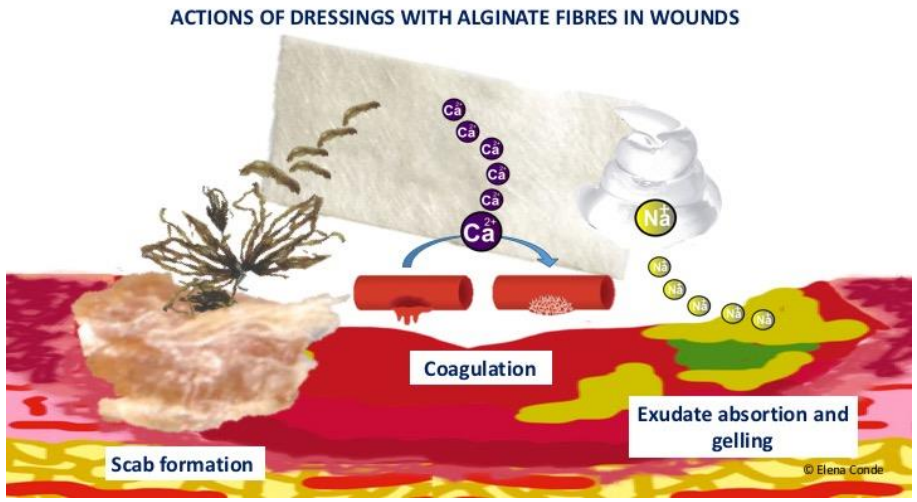
Dental Applications

- Alginate salts are widely used in **dental impression materials**.
- Common alginate salts include:
  - Potassium alginate
  - Sodium alginate
  - Ammonium alginate
- These materials react with **calcium sulfate dihydrate** in water.
- The reaction forms an **insoluble calcium alginate gel**.
- The final impression gel is **elastic** and **irreversible**.



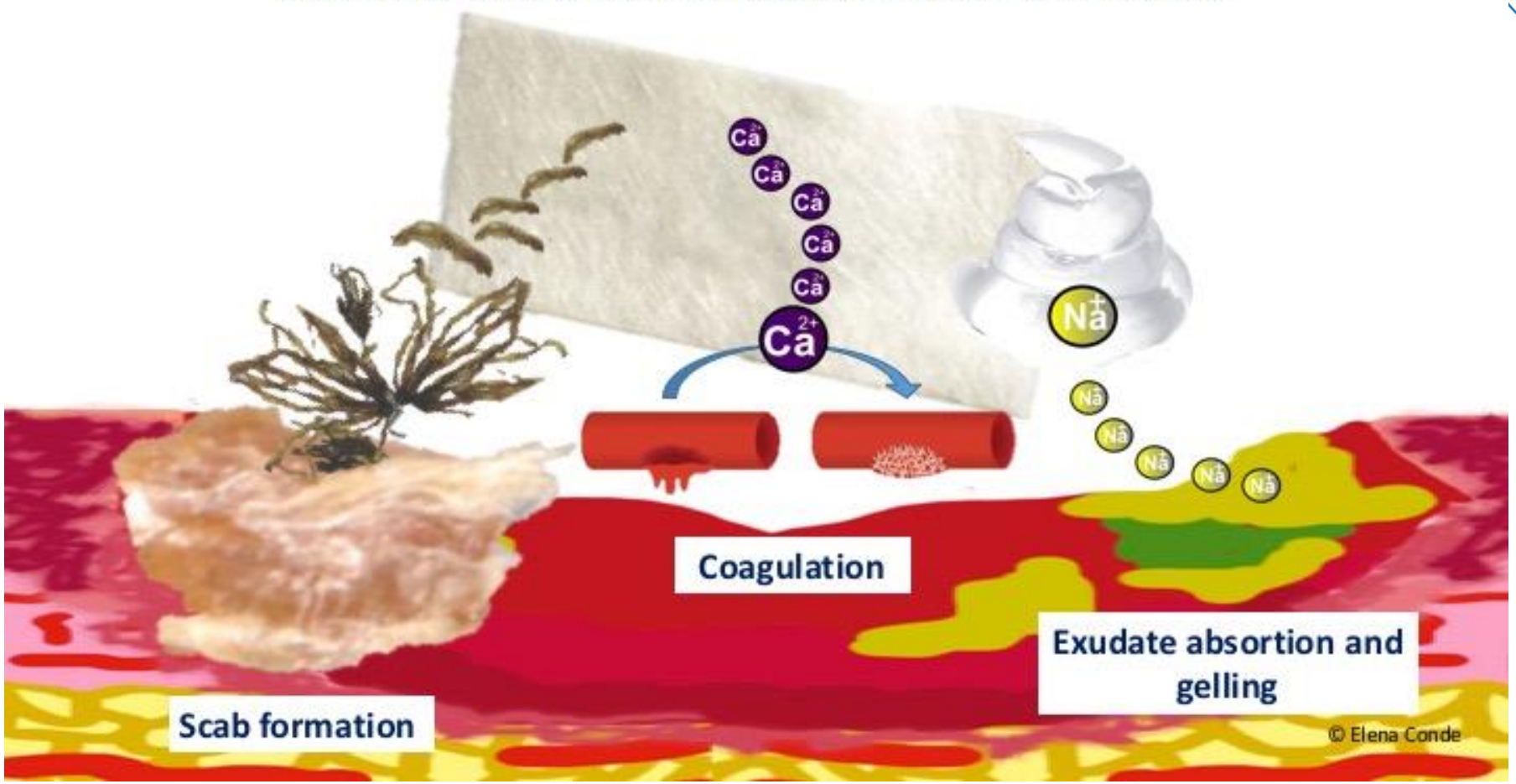
Wound Dressing Applications

- Calcium alginate fibers are used in **wound dressings**.
- They are especially useful for:
  - Large wounds
  - Burns
  - Bleeding wounds



- Alginate dressings can absorb wound fluid.
- They are easier to remove than some cellulose-based bandages.
- This can help reduce disruption of the wound area.

ACTIONS OF DRESSINGS WITH ALGINATE FIBRES IN WOUNDS



Alginate Beads

- Calcium alginate beads can be produced by the **dripping method**.
- Alginate solution is dropped into a calcium-containing solution.
- Beads form almost instantly by ionic cross-linking.
- Beads can range from **microscale** to **millimeter scale**.
- Alginate beads can be used to:
  - Encapsulate cells
  - Immobilize cells
  - Deliver proteins
  - Deliver drugs

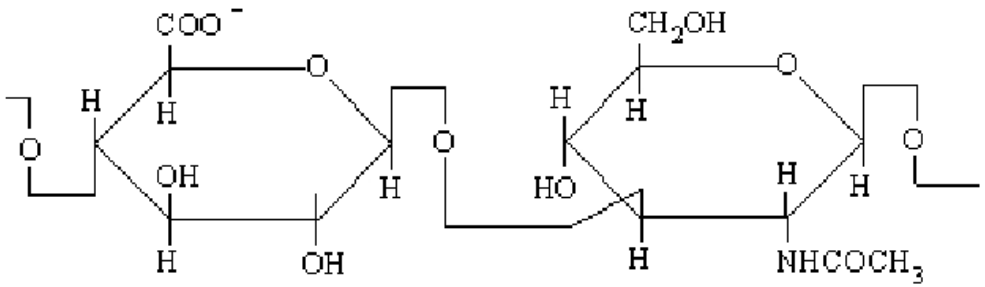
Limitations of Alginate

- Main limitations:
  - Gel degradation can be difficult to control
  - Loss of calcium ions can weaken the gel
  - pH sensitivity can affect release behavior
  - At low pH, alginate may shrink
  - At higher pH, rapid degradation may cause burst release
  - Biomolecules may leak through pores during processing

7.Hyaluronan

Hyaluronan

- Hyaluronan is a natural **polysaccharide-based biomaterial**.
- It is also known as **hyaluronic acid**.
- It is a **linear**, **hydrophilic**, and **polyanionic** polymer.
- Hyaluronan belongs to the group of **glycosaminoglycans**.
- It is widely found in the **extracellular matrix** of the body.
- It is especially important in soft tissues and joint lubrication.



Hyaluronan in the Human Body

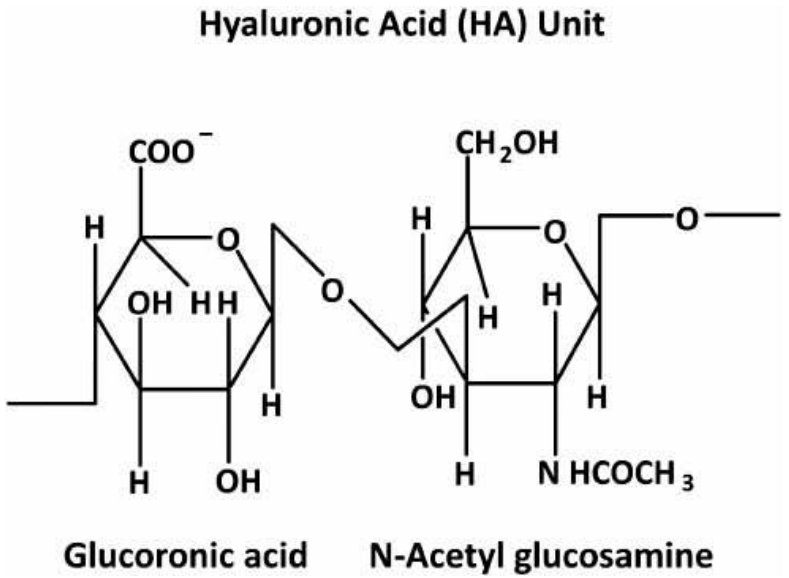
- Hyaluronan is naturally found in:
  - Extracellular matrix
  - Connective tissues
  - Epithelial tissues
  - Neural tissues
  - Synovial fluid
  - Vitreous humor of the eye
  - Umbilical cord



- **The Hyaluronan** is widely distributed in the body, especially in tissues that need hydration, lubrication, or mechanical support.

Basic Structure of Hyaluronan

- Hyaluronan is made of repeating **disaccharide units**.
- Each disaccharide contains:
  - **D-glucuronic acid**
  - **N-acetyl-D-glucosamine**



- These units are linked by alternating **β-1,4** and **β-1,3 glycosidic bonds**.
- The result is a long, linear polysaccharide chain.

Important Physical Behavior

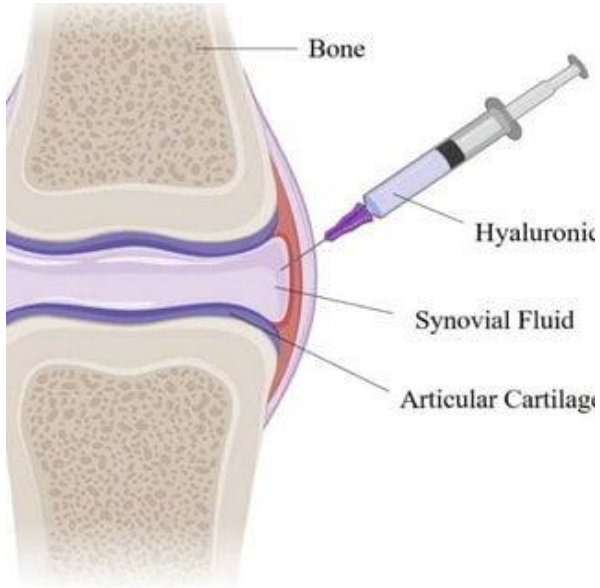
- Hyaluronan is highly **hydrophilic**.
- It can absorb and retain large amounts of water.
- In water, hyaluronan molecules can swell strongly.
- Hyaluronan solutions are highly **viscoelastic**.
- This means they behave partly like a liquid and partly like an elastic material.

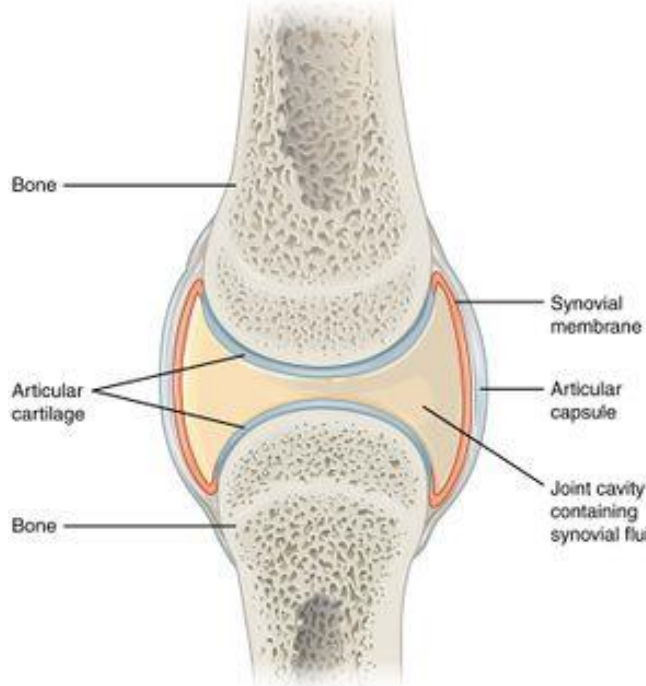
Physiological Roles of Hyaluronan

- Hyaluronan has several important roles in the body:
  - Regulates tissue and matrix water content
  - Supports wound repair
  - Helps tissue regeneration
  - Provides resistance to compression in cartilage
  - Acts as a lubricant in synovial fluid
  - Works as a shock absorber in joints

Hyaluronan in Joints

- Hyaluronan is an important component of **synovial fluid**.
- Its viscoelastic behavior helps joints by:
  - Reducing friction
  - Acting as a lubricant
  - Absorbing mechanical shock
  - Supporting smooth movement





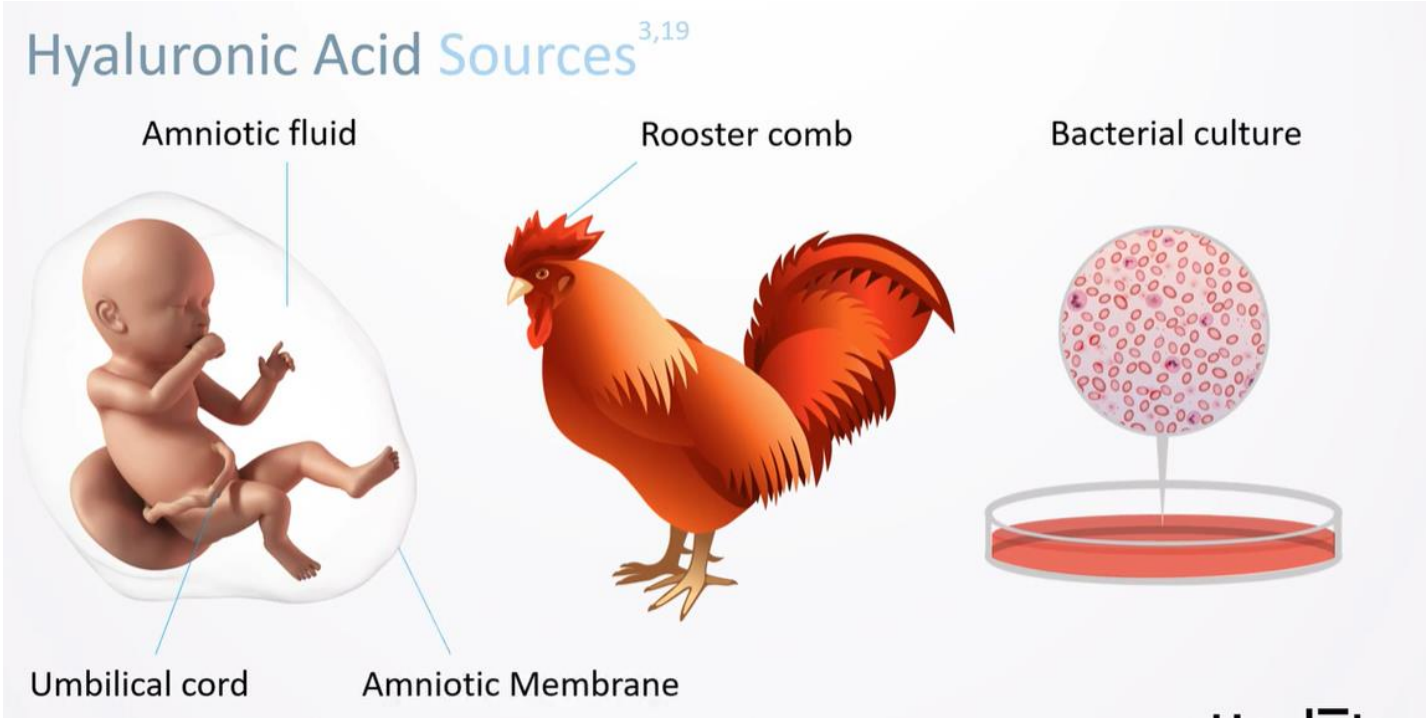
- This is why hyaluronan is important in **orthopedic applications**.

Biological Activity

- Hyaluronan plays an important role in:
  - Tissue regeneration
  - Embryonic development
  - Wound healing
  - Extracellular matrix homeostasis
  - Inflammation regulation
  - Its polyionic structure can help **scavenge free radicals**.
  - This gives hyaluronan an **antioxidant effect**.

Sources of Hyaluronan

- Hyaluronan can be obtained from natural tissues such as:
  - Umbilical cord
  - Rooster comb
  - Synovial fluid
  - Vitreous humor
  - It can also be produced industrially by **microbial fermentation**.



- Some bacterial strains, such as **Streptococci**, can produce hyaluronan.
- It allows large-scale production.

Modification of Hyaluronan

- Hyaluronan has several chemical groups that can be modified:
  - Hydroxyl groups
  - Carboxyl groups
  - Acetamido groups
- Modification is used to control:
  - Solubility
  - Stiffness
  - Degradation rate
  - Biological activity
  - Processing into gels or scaffolds

Derivatization Methods

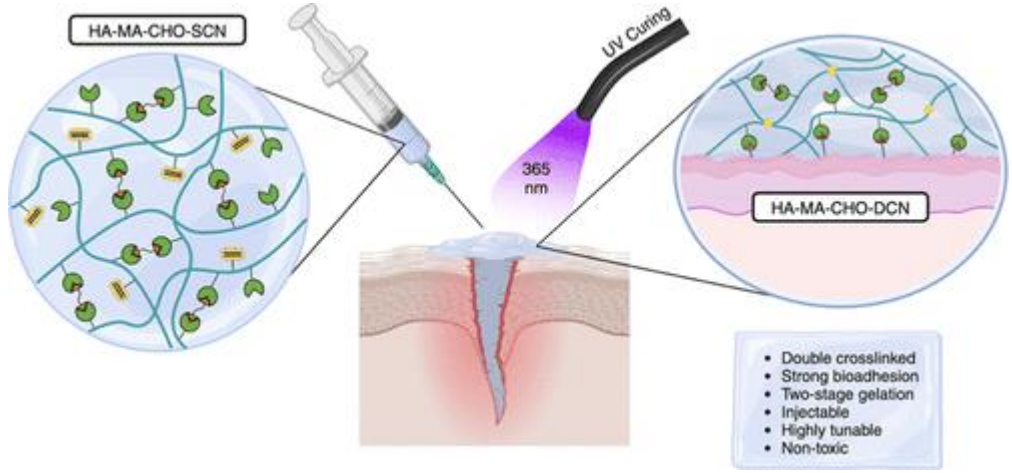
- Hyaluronan can be chemically modified by:
  - **Esterification**
  - **Carbodiimide-mediated modification**
  - **Sulfation**
- Esterification can make hyaluronan:
  - More hydrophobic
  - More rigid
  - Less susceptible to enzymatic degradation
- Sulfation introduces sulfur-containing groups into the structure.

Cross-linking of Hyaluronan

- Cross-linking is used to improve hyaluronan properties.
- Cross-linking can produce stronger and more stable materials.
- Common cross-linking approaches include:
  - Diepoxy cross-linking
  - Carbodiimide-mediated cross-linking
  - Aldehyde cross-linking
  - Divinyl sulfone cross-linking
  - Photocross-linking

Photocross-linking

- Photocross-linking occurs when modified hyaluronan is exposed to a suitable wavelength of light.
- It can occur under **physiological conditions**.
- It is useful because it allows:
  - Hydrogel formation
  - Cell encapsulation
  - Incorporation of bioactive molecules
  - Patterning of hyaluronan surfaces



Biomedical Applications

- Hyaluronan is used in many biomedical fields:
  - Orthopedics
  - Dermatology
  - Ophthalmology
  - Cardiovascular applications
  - Tissue engineering
  - Drug delivery
  - Gene delivery



- It is useful because it is **naturally derived**, **non-immunogenic**, and **modifiable**.

Clinical Uses of Hyaluronan

- In orthopedics:
  - Joint lubrication
  - Mechanical support in osteoarthritis
- In dermatology:
  - Wound dressings
  - Burn treatment
  - Dermal ulcers



- In ophthalmology, it is used as a surgical aid during:
  - Cataract surgery
  - Intraocular lens implantation
  - Glaucoma surgery
  - Retinal surgery
  - Corneal transplant surgery

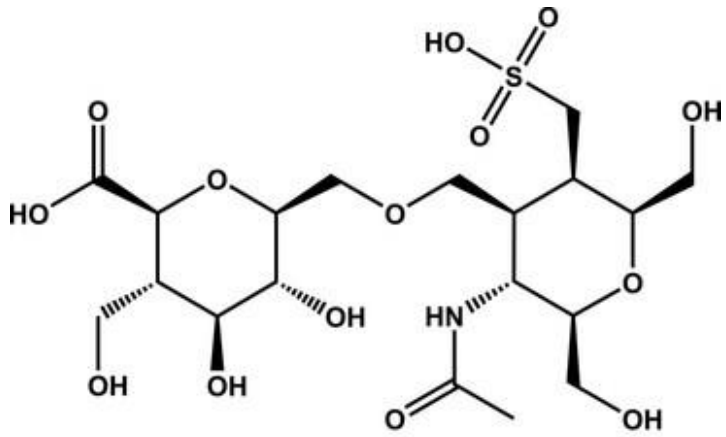
Limitations of Hyaluronan

- Main limitations:
  - Native hyaluronan can degrade quickly
  - Mechanical strength may be limited
  - Properties depend on molecular weight and source
  - Stability often needs improvement by modification
  - Cross-linking may be required for long-term biomedical use

8.Chondroitin Sulfate

Chondroitin Sulfate

- Chondroitin sulfate is a natural **polysaccharide-based biomaterial**.
- It is an **unbranched polysaccharide**.
- It belongs to the group of **sulfated glycosaminoglycans**.
- It is a major component of the **extracellular matrix**.
- It is especially important in tissues such as **cartilage**.



Chondroitin Sulfate

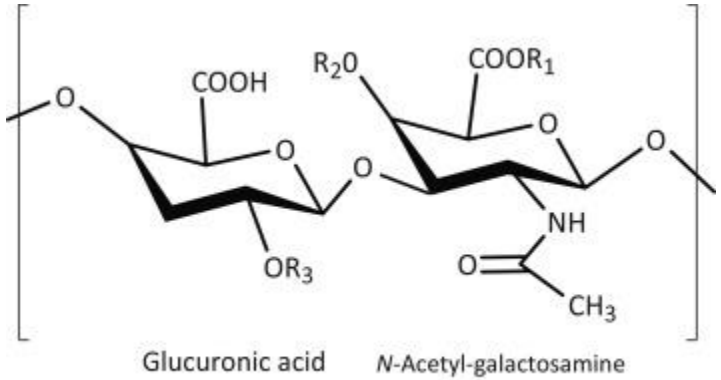
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Role in Cartilage

- Chondroitin sulfate helps maintain the **structural integrity** of cartilage.
- It contributes to the function of the extracellular matrix.
- It helps cartilage resist mechanical loading.
- It is important for tissue organization and biological signaling.
- Its structure allows it to interact with proteins and other matrix molecules.

Basic Structure

- A chondroitin sulfate chain contains more than **100 sugar units**.
- It is made of alternating disaccharides:
  - **N-acetyl-D-galactosamine**
  - **D-glucuronic acid**
- These repeating sugars form a long polysaccharide chain.
- The chain may contain sulfate groups in different positions.

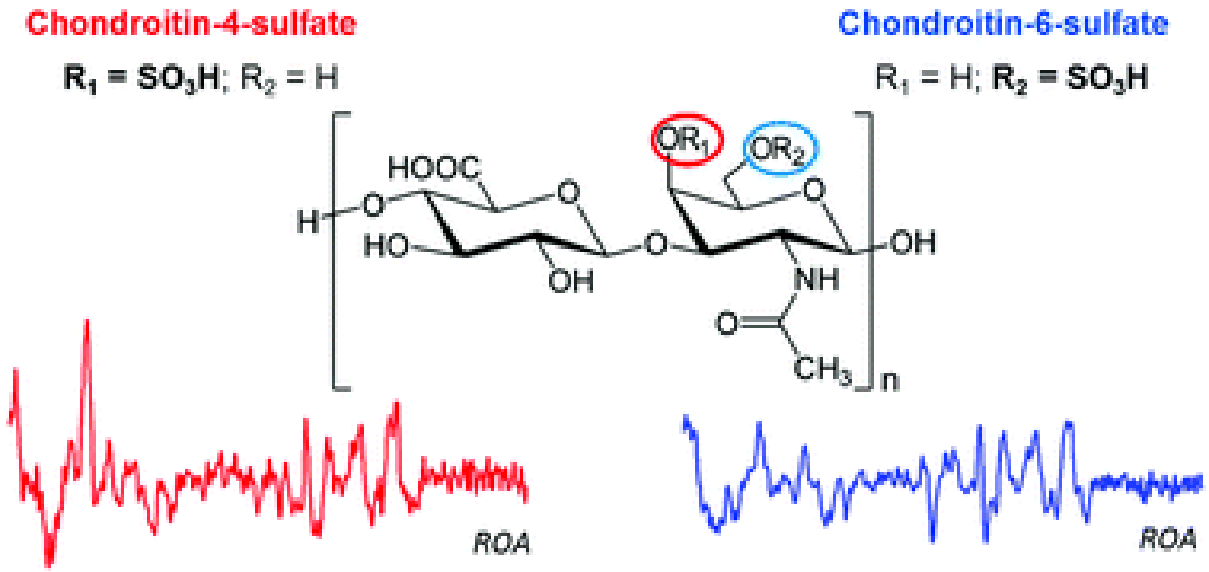


Sulfation

- Chondroitin sulfate contains **sulfate groups**.
- Sulfation usually occurs on **N-acetyl-D-galactosamine**.
- The sulfate group may attach mainly at:
  - Position **4**
  - Position **6**
- The number and position of sulfate groups affect the biological activity of the molecule.

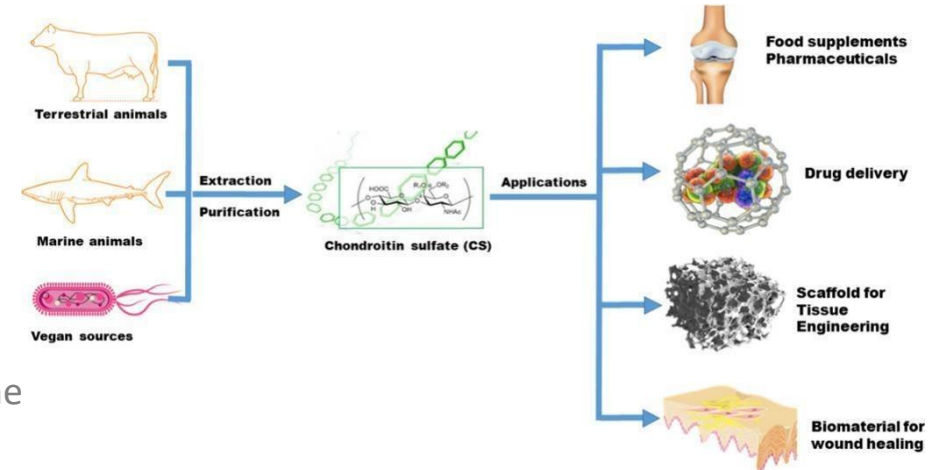
Types of Chondroitin Sulfate

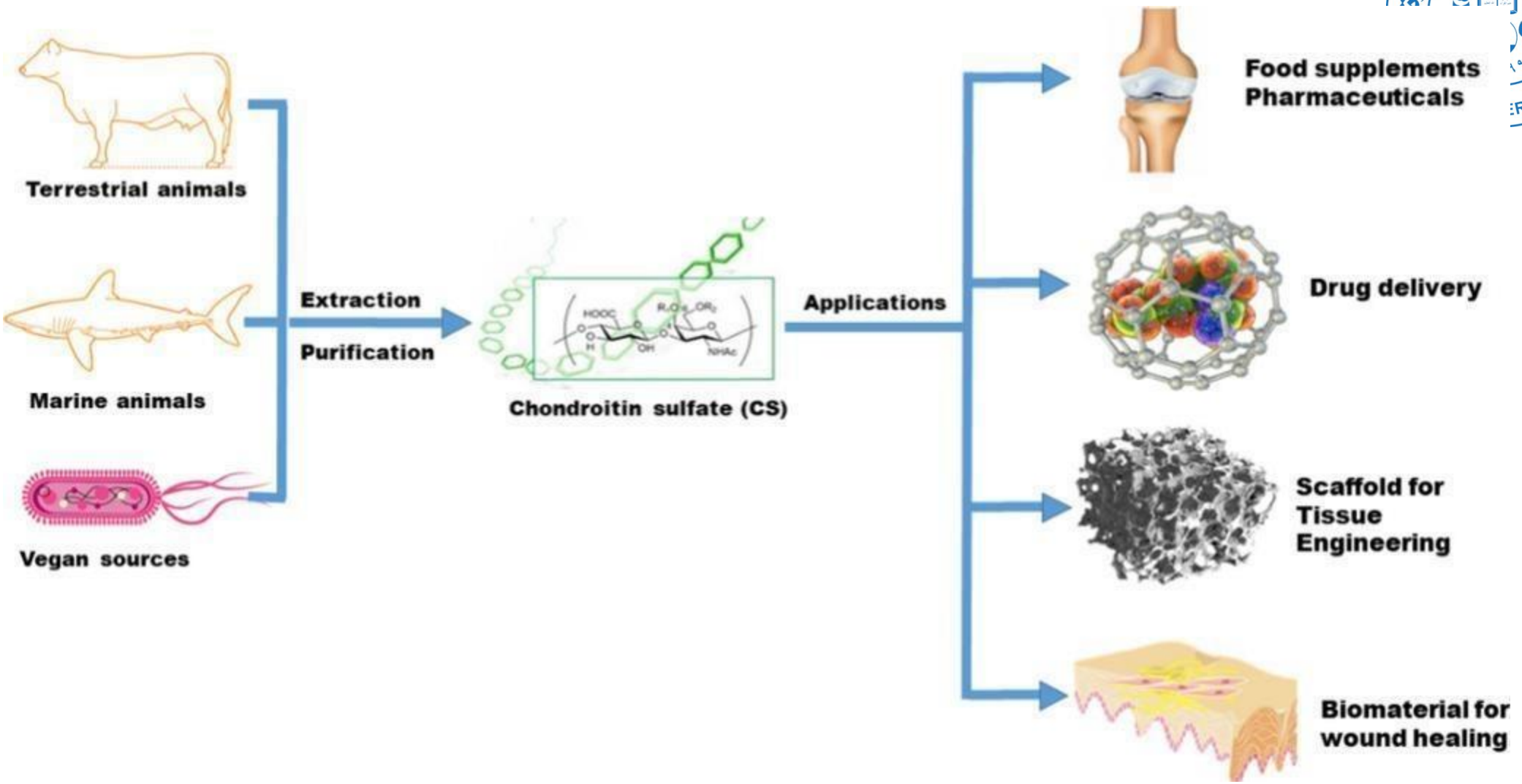
- Two important types are:
  - **Chondroitin 4-sulfate**
    - Also called **chondroitin sulfate type A**
    - Sulfation occurs at position **4**
  - **Chondroitin 6-sulfate**
- Also called **chondroitin sulfate type C**
    - Sulfation occurs at position **6**



Sources of Chondroitin Sulfate

- Chondroitin sulfate is commonly obtained from animal tissues.
  - Bovine nasal septum
  - Bovine trachea
  - Shark cartilage
- It is produced by **enzymatic digestion** of these tissues.
- Marine and bovine sources are commonly used commercially.




Terrestrial animals
Marine animals
Vegan sources
Extraction
Purification
Chondroitin sulfate (CS)
Applications
Food supplements
Pharmaceuticals
Drug delivery
Scaffold for
Tissue
Engineering
Biomaterial for
wound healing

Biological Properties

- Chondroitin sulfate has several useful biological properties:
  - Can bind and regulate **growth factors**
  - Can modulate **cytokines**
  - Can inhibit some **proteases**
  - Supports cell adhesion
  - Supports cell migration
  - Supports cell proliferation and differentiation
  - Non-immunogenic
  - Degrades into non-toxic oligosaccharides

Biomedical Applications

- Chondroitin sulfate is used in several biomedical applications:
  - Nutritional supplements for osteoarthritis
  - Tissue engineering scaffolds
  - Skin regeneration products
  - Burn treatment
  - Reconstructive surgery
  - Ophthalmic surgical aids
  - Drug and growth factor delivery

Commercial Examples

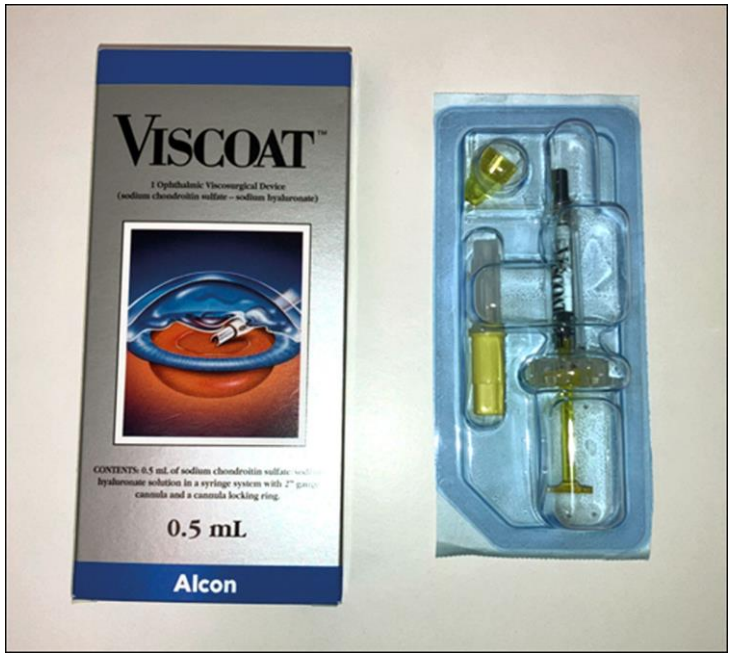
- Integra® Dermal Regeneration Template
  - Contains chondroitin sulfate with cross-linked collagen
  - Used for burns and reconstructive surgery

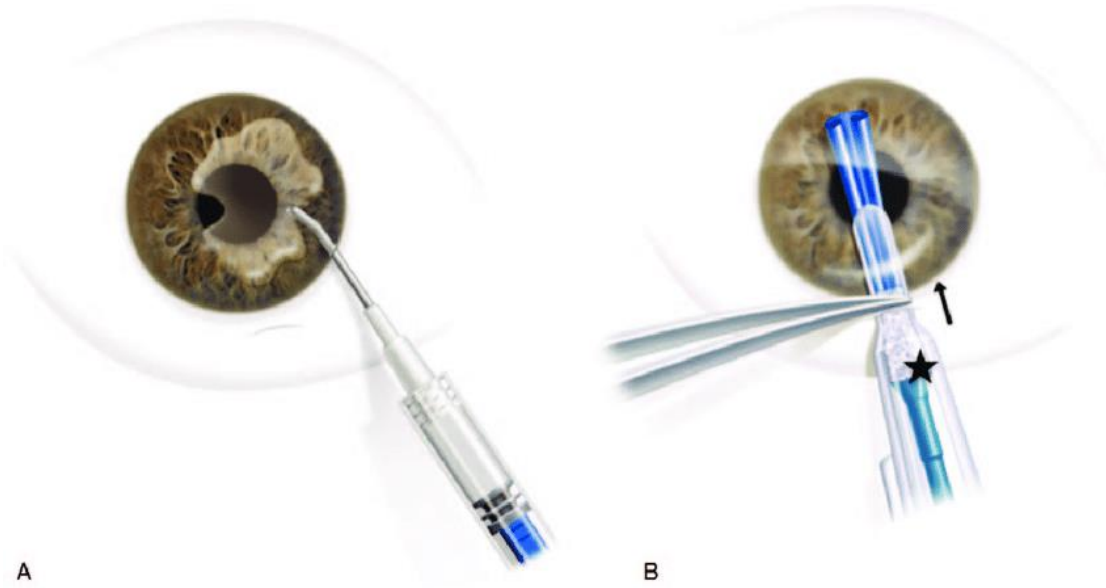




Commercial Examples

- Viscoat® Ophthalmic Viscoelastic Device
  - Contains chondroitin sulfate with sodium hyaluronate
  - Used in eye surgery to protect the corneal endothelium





Limitations

- Main limitations:
  - Usually needs to be combined with other polymers for stronger structures
  - Properties depend on source and sulfation pattern
  - Animal-derived sources may show variability
  - Mechanical strength alone is limited
  - Cross-linking or composite formation may be needed for scaffold applications

9.Coral

Coral

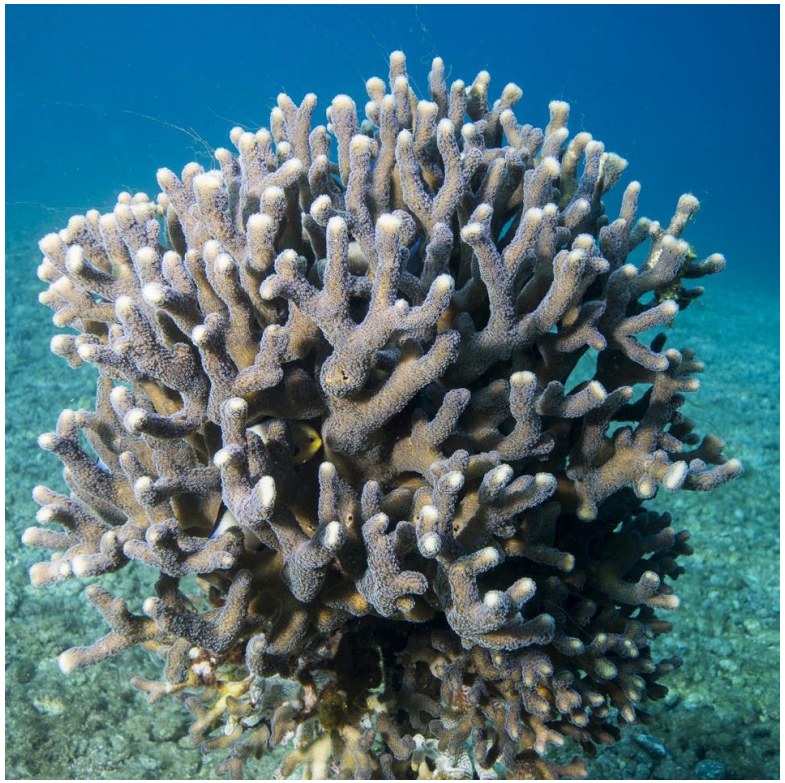


- Natural coral has been investigated as a **bone substitute material**.
- It has been used clinically for:
  - Filling bone defects
  - Onlay grafts
  - Orthopedic and traumatology applications
  - Craniofacial, dental, neurosurgical, and spinal fusion applications
- Coral is useful because its porous structure resembles **cancellous bone**.
- **Notes:**
  Introduce coral as a natural non-polymeric biomaterial. Unlike collagen, chitosan, or cellulose, coral is mainly used as a mineral-based bone graft substitute.

Source of Medical Coral

- Medical coral grafts are derived from the **exoskeleton of stony corals**.
- These corals belong mainly to reef-building corals.
- Important coral genera include:
  - Porites
  - Goniopora
  - Acropora
  - Madrepora
  - Lobophyllia
  - Polyphyllia
  - Pocillopora





Basic Composition of Coral

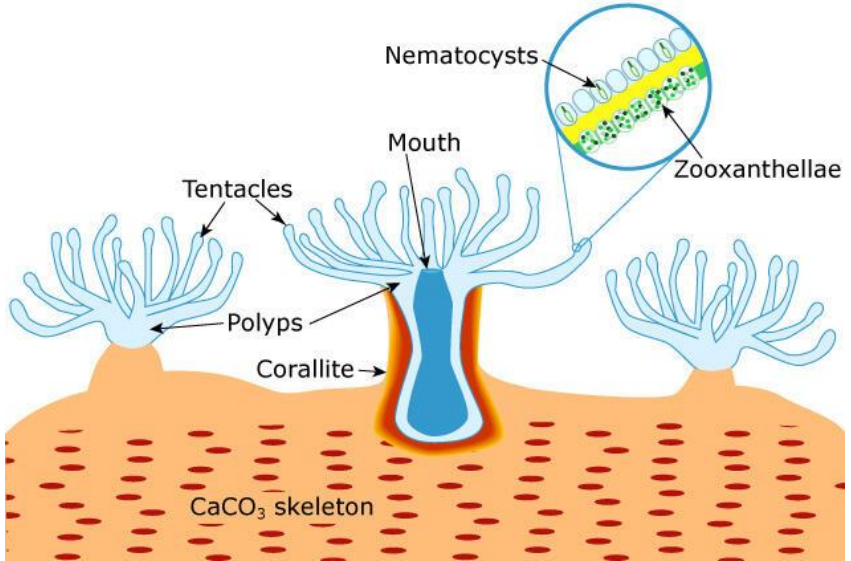
- Coral skeleton is mainly made of **calcium carbonate ($CaCO_3$)**.
- The mineral form is mostly **aragonite**.
- Coral may also contain small impurities such as:
  - Magnesium
  - Potassium
  - Sodium
  - Amino acids
  - Trace strontium, fluorine, and phosphorus
- These minor components can help in the **calcification process**.



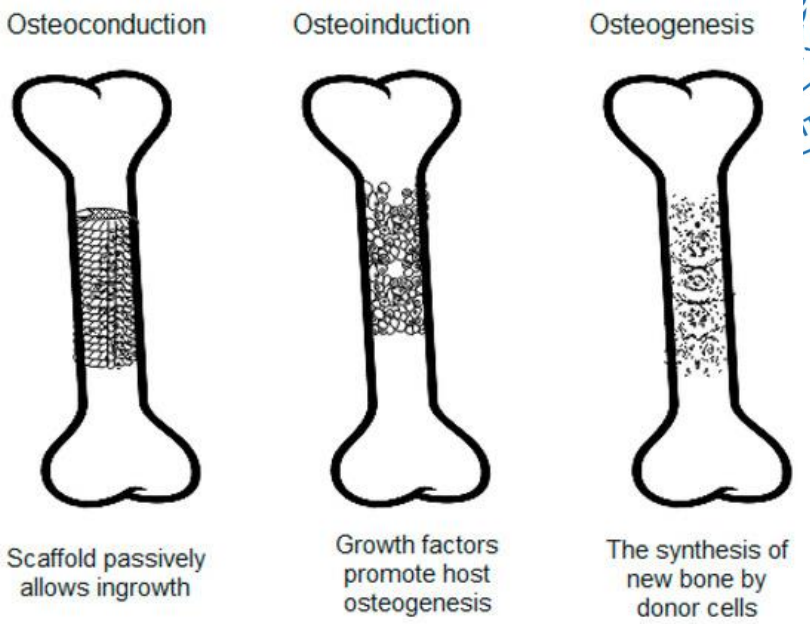
Coral Structure



- Coral is produced by colonies of small organisms called **polyps**.
- The outer coral layer contains cells that secrete the mineral phase.
- After calcification, the secreted calcium carbonate forms a porous coral scaffold.
- The scaffold architecture depends on the **coral species**.



Why Coral Is Useful for Bone Repair



- Natural coral is:
  - **Biocompatible**
  - **Osteoconductive**
  - Porous
  - Similar to cancellous bone in structure
- Its mechanical properties are comparable to **cancellous bone**.
- The porous structure allows bone tissue to grow into the material.
- *About Osteoconduction. Coral does not directly create bone by itself, but it supports bone growth along its structure.*

Effect of Pore Interconnectivity

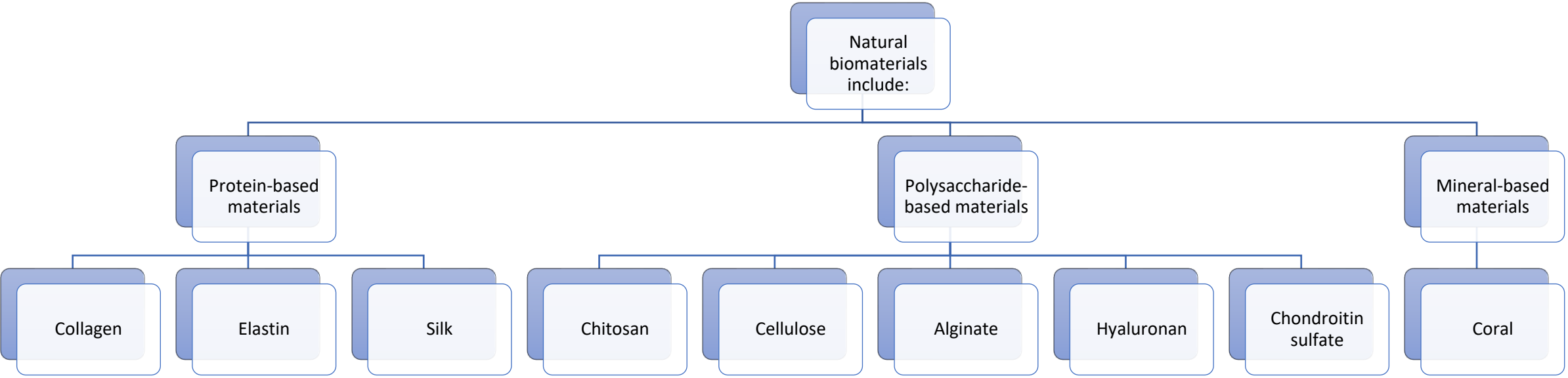
- Highly interconnected pores improve:
  - Bone ingrowth
  - Nutrient diffusion
  - Vascularization
  - Resorption behavior
- Poorly interconnected pores can reduce resorption and slow bone formation.
- Some corals, such as **Polyphyllia**, have incomplete pore interconnection.
- This can make them resorb more slowly and provide better **volumetric stability**.

Limitations of Coral

- Coral has some important limitations:
  - Brittleness
  - Limited strength for load-bearing applications
  - Variability between coral species
  - Inconsistent degradation after processing
  - Possible instability if untreated
  - Dependence on pore size and interconnectivity
- These limitations mean coral is mainly suitable for selected bone graft applications.

Summary

The main classification



Advantages of Natural Biomaterials

- Natural biomaterials are attractive because they often have:
  - Good biocompatibility
  - Biodegradability
  - Low cost
  - Availability from natural sources
  - Similarity to extracellular matrix components
  - Ability to interact with cells
- These properties make them useful in many biomedical applications.
- Natural materials are attractive because the body can often recognize or tolerate them better than fully synthetic materials.

Limitations of Natural Biomaterials

- Natural biomaterials also have some limitations:
  - Batch-to-batch variability
  - Limited mechanical strength
  - Limited processability
  - Possible rapid degradation
  - Possible swelling in physiological fluids
  - Properties depend strongly on source and processing
- These limitations are often improved by **cross-linking**, **chemical modification**, or **composite formation**.

Think with me 🤔🤔🤔

- For each natural biomaterial mentioned in the lecture, write:
  - 1. Its main natural source(s)
  - 2. The main biomedical application



Thank you